Estimator or experiment? A Cramér–Rao accounting of parameter recovery by physics-informed and hybrid neural–mechanistic models of WNT–retinoic acid–HOX signalling in colorectal cancer

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Abstract

Physics-informed neural networks (PINNs) and neural–mechanistic hybrid models estimate the parameters of signalling models, and their failures are attributed either to the estimator or to non-identifiability. They call for opposite remedies and are rarely told apart. The Cramér–Rao bound of the experiment performed separates them. We carry this accounting through a seven-state model of WNT, retinoic acid and HOX signalling in colorectal cancer, from healthy crypt to severe APC loss. Three inverse PINN formulations recover 2–21 of 34 parameters per regime, but the bound says an efficient estimator would recover 132.5 of 136 regime–parameter pairs, a converged maximum-likelihood fit attains it on independent datasets where the data are informative, and the PINNs' errors are 10–92 times the bound at the median: an estimator gap. A Schur-complement Gauss–Newton PINN, which freezes the network's features so that the problem becomes separable with a block-arrowhead structure, closes the gap wherever its continuation converges (both low-WNT regimes on two datasets), there matching the classical estimator and reporting its own precision. Where the bound itself is wide no estimator helps: APC functionality, the parameter that defines the disease stage, loses three orders of magnitude of precision (the identifiable product it enters, 28-fold) as β-catenin saturates, and one WNT knockdown restores it while information-matched conditions do not. A learned regulatory term opens an analogous gap: its constraint $f(0) = 0$ is informative only if a condition drives the regulator to zero, so the host equation's basal rate is lost under every network parameterisation tried. A Fisher analysis of the hybrid equation, a fixed-effort dose-response and a prospective test of a training-free design table show that a knockdown visiting the anchor, in a condition that still contains the basal term, removes most of the loss. The first diagnostic should be the design's information budget.

Author summary

Mathematical models of cell signalling are only as useful as the numbers that go into them, and those numbers, rates of production and degradation, are usually estimated from experiments. Machine-learning methods that combine neural networks with the model's equations promise to do this from sparse data, but when an estimate comes out wrong it is hard to know why: was the method too weak, or did the experiment simply not contain the information? We show that a classical quantity from statistics, the Cramér–Rao bound, answers this question before any network is trained, and we use it to work through a model of the signalling network that goes awry in colorectal cancer. For most parameters the information was there and a better solver, which we develop, recovers them. For the parameter that marks the stage of the disease, and for parameters that sit next to a term learned by a network, the information was missing, and only a different experiment, a targeted knockdown, brought it back. The bound tells an experimentalist which of the two to change.

1 Introduction

Mechanistic differential-equation models of signalling are useful in proportion to what their parameters mean. In a model of the colonic crypt, a rate constant for APC-dependent degradation of β-catenin is not a fitting coefficient: it is the quantity that separates a healthy crypt from an adenoma, and estimating it from data is the point of the exercise [1–3]. Two machine-learning ideas have changed how such estimates are attempted. Physics-informed neural networks (PINNs) recover parameters by training a network to fit the data while satisfying the differential equations [4, 5], and have been applied to systems-biology inverse problems from sparse, noisy measurements [6–9]. Universal differential equations

and neural–mechanistic hybrids keep the trusted parts of a model and learn the untrusted part, typically one regulatory law, with a small embedded network [10–13].

Both literatures report parameter recovery the same way: a table of estimates, a count of how many fall within some tolerance of the truth on synthetic data, and, for the ones that do not, an attribution. The attribution is usually one of two kinds. Either the parameter is "non-identifiable" [14, 15], or the estimator struggled: PINN training is known to be stiff, to balance its loss terms badly, and to stall [16–18], and in hybrid models the over-parameterised network component is reported to have "consistently hindered parameter identifiability" [11]. The two attributions call for opposite remedies. An estimator failure is fixed by a better estimator; an information failure is fixed only by a better experiment, and no amount of estimator engineering will touch it. Yet the two are routinely diagnosed by inspection, and an improvement in the count from a new architecture is indistinguishable, in such a table, from an improvement that a new experimental condition would have given for free.

This paper argues that the question *estimator or experiment?* has a quantitative answer that costs one sensitivity computation, and that asking it systematically changes the conclusions one draws. The answer is the Cramér–Rao lower bound (CRLB) [19, 20] of the experiment actually performed: the Fisher information of the design at the true parameters, inverted. It says, before any estimator is trained, how precisely every parameter *can* be known from these data. It is local and asymptotic, so we verify it empirically; but where it holds it splits every recovery failure into two parts that we call the *estimator gap*, the distance between what an estimator achieves and what the bound permits, and the *information gap*, the parameters the bound itself declares unknowable. The first is closed by numerics, the second only by design [21–23].

We carry this accounting through a single model at increasing levels of difficulty, from the forward problem to the inverse problem to the hybrid one. The model is a reduced seven-state description of the WNT, retinoic acid (RA) and HOX antagonism in the colonic crypt with 36 kinetic parameters, studied in four regimes from a healthy crypt to severe APC loss and observed under a ten-condition design of retinoid and WNT/MYC perturbations (Fig 1). Four findings organise the paper.

1. *The PINN's recovery shortfall is an estimator gap.* The best of three matched PINN variants recovers 21, 19, 10 and 10 of the 34 regularly behaved parameters in the four regimes, a count that reads like a hard limit of the data. The bound says otherwise: an efficient estimator would recover 132.5 of the 136 regime–parameter pairs outside one structural degeneracy, a converged maximum-likelihood fit attains the bound's standard deviations on independent datasets where the data are informative and never does worse, and on the benchmark data, given a few starts, it recovers every one of them.

2. *The estimator gap of a PINN can be closed without leaving the PINN framework.* Freezing the network's hidden features and optimising only its output layer together with the parameters turns the inverse problem into a separable least-squares problem whose Gauss–Newton matrix is block-arrowhead. A Schur-complement Gauss–Newton PINN (SGN-PINN) built on this matches the classical estimator parameter by parameter wherever its continuation converges, which on two datasets includes both low-WNT regimes and, on the held-out one, Advanced Adenoma, and it reports an estimate of its own Cramér–Rao precision as a by-product; where the continuation stalls, its physics residual says so.

3. *The parameter that defines disease stage is the one the design does not inform.* APC functionality θ_P enters only through a product with a degradation constant; with that structural degeneracy removed, its bound still widens from 0.04% to 57% relative standard deviation (that of the identifiable product it enters, 28-fold) as WNT drive saturates β-catenin. No estimator can recover what the bound withholds; one added WNT knockdown restores it, and two added conditions that perturb without leaving the saturated plateau do not.

4. *A learned regulatory term creates an information gap of its own.* When a flexible network replaces a production term $a + f(u)$, the constraint $f(0) = 0$ that is meant to separate the basal rate a from the learned function carries information only if the experiment drives the regulator u near zero, which no condition of a conventional design does. The basal rate is then lost under every network parameterisation we tried. A Fisher-information analysis of the hybrid equation, consistent with fits run before it, shows that two added conditions that do not visit the anchor leave the loss unchanged while a depletion knockdown that visits it removes most of the loss; and a training-free design table built on the idea predicted the right knockdown prospectively, including one informative failure that the Fisher analysis explains.

The thread through these findings is that a recovery count, by itself, does not say whether to change the estimator or the experiment, and the bound does. The same yardstick that exonerates the data in the first two findings indicts them

in the last two, and in both cases it names the remedy: a solver where the gap is numerical, a perturbation that visits the region in which the parameter acts where the gap is informational. We frame the latter as a partial-persistence-of-excitation condition [24, 25] applied to a model constraint rather than to an estimate. All data are synthetic, so that truth is known and every claim can be checked; the limitations this imposes are discussed in Section 3.

2 Results

2.1 A seven-state model of the crypt, four disease regimes and a ten-condition design

The model couples three loops (Fig 1A; Methods, Eqs (2)–(8)): a pro-stemness positive loop β-catenin→MYC→ HOXA13→β-catenin; a differentiation loop in which RA induces HOXA5, HOXA5 supports APC and APC degrades β-catenin; and RA homeostasis through its catabolising enzyme CYP26A1, which β-catenin also induces. Disease progression is a coupled sweep that raises WNT drive W while lowering APC functionality θ_P (Table 1); every other parameter is shared. Along the sweep the β-catenin level roughly triples, APC falls to a sixth of its healthy level and the stemness index rises fourfold (Fig 1B). The experiment that every inverse problem in this paper sees is the ten-condition design of Fig 1C: six conditions vary only the retinoid input and four perturb the WNT/MYC sub-network directly. All seven states are observed at 150 times per condition with Gaussian noise of standard deviation 0.002, 0.1–0.7% of the mean level of each state, except APC, whose level falls with WNT drive (0.4% in Normal to 2.6% in Severe APC Loss). This is a generous design, about 10^4 observations per regime; how the conclusions depend on it is examined in S1 Fig.

2.2 The forward problem is not the main bottleneck

Before asking how well a PINN estimates parameters, we ask whether it can represent the dynamics at all, because an inverse failure could simply be a representation failure. With the parameters known and 40 noise-free observations per state, a Fourier-feature PINN reproduces all seven states in all four regimes (Fig 2A) with a mean relative L^2 error of 2.83%, 1.76%, 2.09% and 2.95% (grand mean 2.41%, or 1.59% without APC), against 1.06% for the same network trained on 100 evenly spaced labels (Fig 2B). The largest relative error is in APC in the high-WNT regimes, where APC settles near zero and the denominator of the relative error is small (13.5% relative, 1.3% normalised RMSE). The network is expressive enough for the trajectories, including the circadian oscillation and the treatment pulse. Its error is larger than the 0.1–2.6% observation noise of the inverse problem, so representation is not free, but it is a few percent where the parameter errors of the inverse PINN below are tens of percent; the representation of the states is not what limits them (Section 2.6 returns to this).

2.3 The information budget of the design

The Fisher information of the ten-condition design at the true parameters, Eq (9), is a single Jacobian computation that needs no estimator. In every regime its 35 informative eigenvalues span 5.7 to 8.3 orders of magnitude, and one further eigenvalue sits 7.8 to 10.9 orders below them, at the level of finite-difference round-off: a structural null direction (Fig 3A). The null direction is exactly the analytic degeneracy of Eq (3): it lies in the $(\log\theta_P, \log\delta_{P1})$ plane to within 10^{-6}, with loadings that match the tangent of the level set $\delta_{P1}(1-\theta_P) = \text{const}$ (Fig 3B). Its θ_P loading rises from 0 to 0.95 across the regimes, but that is algebra, the ratio $(1-\theta_P) : \theta_P$, not a sign of saturation; we mention it because it is easily mistaken for one. The Fisher matrix finds the one structural fact about the model without being told, which we take as a check of the computation.

With δ_{P1} known, the CRLB of the remaining 35 parameters is the information budget of the design (Fig 3C). In the two low-WNT regimes every parameter has a relative standard deviation below 1.3%; the budget then shrinks with WNT drive, to 31 of 35 parameters below 5% in Advanced Adenoma and 25 of 35 in Severe APC Loss, where θ_P (57%), ρ_{13} (21%), ε_P (18%) and a_C (16%) are the least informed. Translated into the currency the literature on this model has used, an efficient estimator would recover on average 34.0, 34.0, 33.4 and 31.1 of the 34 parameters outside the degenerate pair to within 10% (the CRLB-expected count $\mathbb{E}[N_{10}]$, Methods), 132.5 of 136 in total (Fig 3D). A naive reading of recovery counts, of the kind the PINN results below produce, would conclude that the data support a dozen or so parameters per regime. The bound says the data support nearly all of them.

2.4 The bound is attained

A Cramér–Rao bound is a statement about a hypothetical efficient estimator in the asymptotic limit; it becomes a yardstick only if some estimator actually attains it on this problem. We tested this directly. For each regime, 24 independent datasets (data seeds 101–124, the same design and noise level) were fitted by the exact maximum-likelihood estimator with δ_{P1} known, started at the truth so that the question is statistical rather than one of finding the basin (Methods). In the two low-WNT regimes the bound is attained: the empirical standard deviation of every log-parameter across datasets

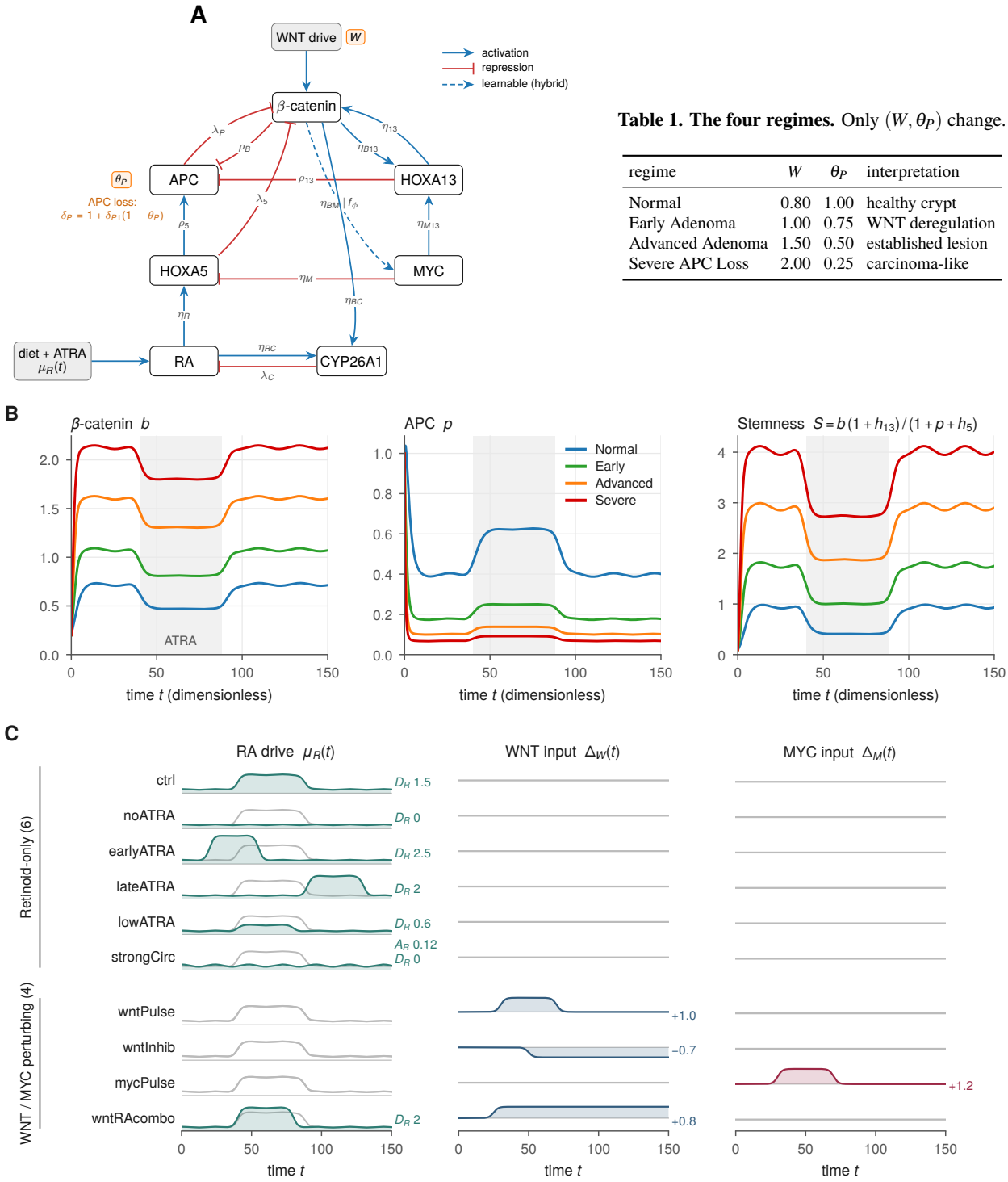


Table 1. The four regimes. Only (W, θ_P) change.

regime	W	θ_P	interpretation
Normal	0.80	1.00	healthy crypt
Early Adenoma	1.00	0.75	WNT deregulation
Advanced Adenoma	1.50	0.50	established lesion
Severe APC Loss	2.00	0.25	carcinoma-like

Fig 1. Model, regimes and experimental design. (A) The WNT–RA–HOX network: one edge per term of Eqs (2)–(8), labelled with its parameter; blue arrows activate, red bars repress. Orange badges mark the two parameters that define the disease regimes; APC loss acts through the excess degradation δ_P. The dashed edge is the main learned term of the hybrid models. (B) Reference trajectories (Radau, tolerance 10^{-10}) of β-catenin, APC and the stemness index S under the control protocol in the four regimes; grey band, ATRA pulse. (C) The ten conditions of the base design: per condition, the retinoid drive $\mu_R(t)$, the exogenous WNT input and the exogenous MYC input, coloured where they differ from the control protocol (grey) and labelled with their amplitudes.

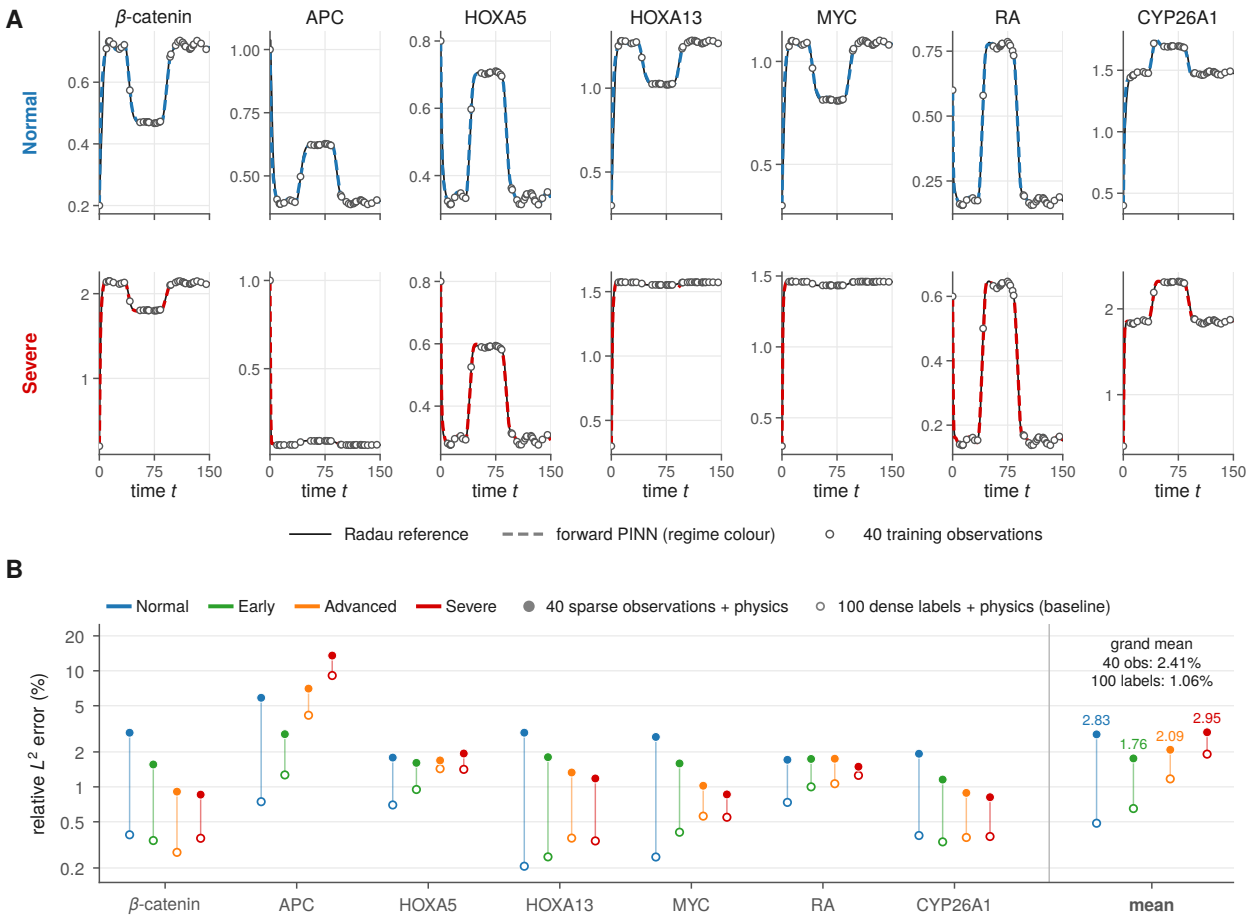


Fig 2. A PINN represents the dynamics. (A) Forward PINN (dashed, regime colour) trained on 40 noise-free observations (open circles), the initial condition and the ODE residual, against the Radau reference (black), for the seven states in the Normal and Severe APC Loss regimes. (B) Relative L^2 error per state and regime of this sparse-data PINN (filled) and of the same network trained on 100 evenly spaced labels (open). Mean errors 2.83/1.76/2.09/2.95% (grand mean 2.41%, against 1.06%); the APC error in the high-WNT regimes is large in relative terms because APC approaches zero there.

matches its CRLB, from 2×10^{-4} to 10^{-2} (Fig 4A), the standardised errors are standard normal (variance 1.03 and 1.13; Fig 4B), the nominal 95% intervals $\pm 1.96 s_j$ cover the truth in 95% and 92% of cases (Fig 4C), and twice the drop in cost from the truth to the estimate follows χ^2_{35} (mean 36.3 and 33.1; Fig 4D). In the two high-WNT regimes the estimates scatter less than the bound (variance 0.88 and 0.70, coverage 95% and 97%) and the likelihood-ratio statistic falls short of χ^2_{35} (mean 31.9 and 31.0). This is not a failure to converge: restarting six fits per regime from their own estimates with stricter stopping rules lowers the cost by at most 2×10^{-4}, against a shortfall of about four units in the statistic, and moves no log-parameter by more than 0.003, and tightening the solver moved the likelihood-ratio statistic by at most two units. It is the non-asymptotic regime of a weakly informed likelihood: along the sloppiest directions the log-likelihood is far from quadratic over the scale of the noise, the estimator is biased there, and its scatter can fall below a bound that holds for unbiased estimators (the weakest parameter, θ_P in Severe APC Loss, scatters by 33% against a bound of 57%). The bound is therefore attained where the data are informative and conservative where they are not; in neither case does a converged estimator do worse than it, which is what a yardstick for estimator performance requires.

2.5 Physics-informed estimators leave an estimator gap

With the yardstick established, the inverse PINN can be placed against it. Fig 5A compares, on the benchmark data and on an independent replicate dataset, three PINN variants (each trained once from the common starting guess) with the classical fit and the bound: the standard formulation with an automatic-differentiation residual, the same with relative state weights and a fixed collocation grid, and the same with the derivative-free integral residual of Eq (10) (Methods).

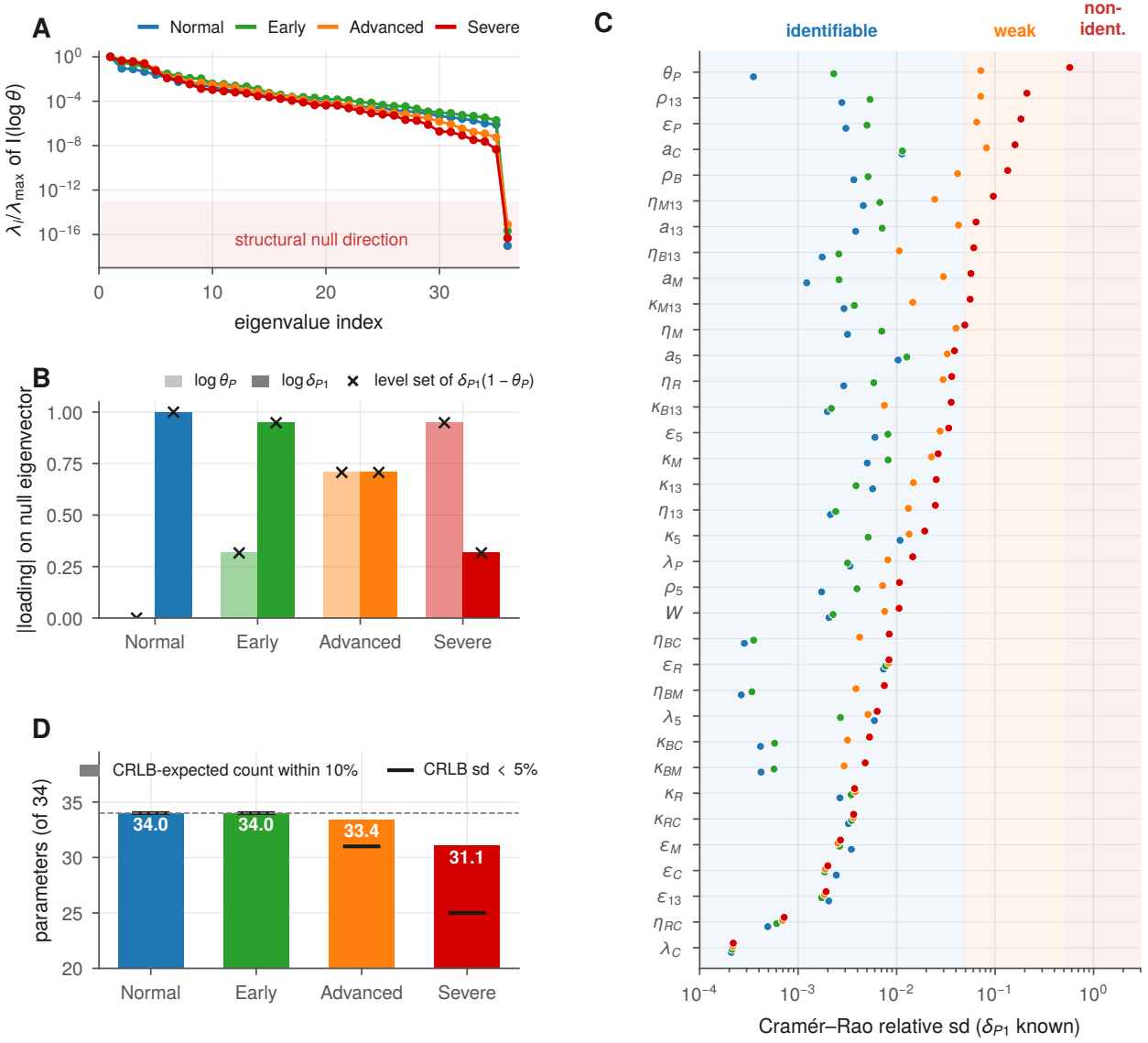


Fig 3. The information budget of the ten-condition design. Fisher information of the 36 log-parameters at the truth (150 observations per condition, seven states, $\sigma = 0.002$). (A) Eigenvalue spectra normalised by the largest eigenvalue: one structural null direction in every regime. (B) Loadings of the null eigenvector on $\log \theta_P$ (light) and $\log \delta_{P1}$ (dark); crosses, the tangent of the level set $\delta_{P1}(1-\theta_P) = \text{const}$, which they match exactly. (C) Cramér–Rao relative standard deviation of each parameter with δ_{P1} known, sorted by its value in Severe APC Loss; background bands, identifiable (< 5%), weak (5–50%), non-identifiable (> 50%). (D) CRLB-expected number of the 34 parameters outside the degenerate pair that an efficient estimator recovers within 10% (bars), and the number with CRLB standard deviation below 5% (ticks).

On the benchmark data the autodiff PINN recovers 14, 10, 7 and 2 of the 34 regular parameters, with median errors 43 to 92 times the bound. Relative state weights and a fixed collocation grid lift this to 21, 19, 10 and 10 (10 to 22 times the bound). Replacing the residual by the integral form on top of them gives 17, 15, 10 and 7 (13 to 33 times), so on these data, at one start each, the derivative-free residual adds nothing to the weighting. No formulation comes near the bound (Fig 5A,C). The integral-residual PINN, the starting point of the SGN-PINN below, recovers 17, 15, 10 and 7 of the 34 regular parameters in the four regimes on the benchmark data (18, 16, 13 and 9 on the replicate), with median relative errors of 8.5%, 12.2%, 15.6% and 30.6%. On the same data the converged classical fit recovers all 34 in every regime, and the CRLB-expected count is 34.0, 34.0, 33.4 and 31.1. Its errors sit at or below the bound: the median ratio of error to CRLB standard deviation is 1.11, 0.60, 0.21 and 0.42 (an efficient estimator would show 0.67, the median of

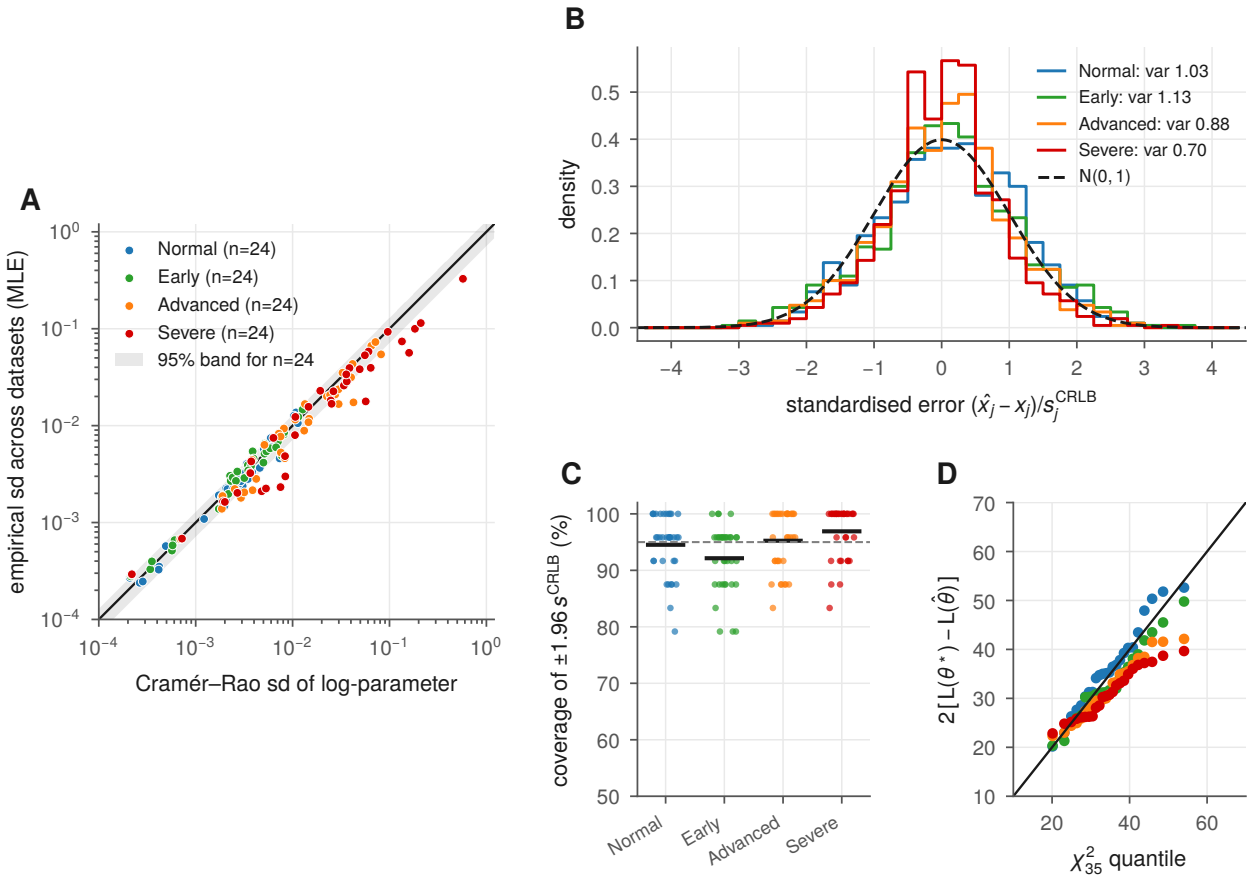


Fig 4. The Cramér–Rao bound is attained where the data are informative. Exact maximum-likelihood estimates (δ_{P1} known, started at the truth) on 24 independent datasets per regime. (A) Empirical standard deviation of each log-parameter across datasets against its CRLB; grey band, the 95% sampling interval of a standard deviation estimated from 24 draws. (B) Standardised errors $(\hat{x}_j - x_j)/s_j$ pooled over parameters and datasets, against the standard normal density (dashed). (C) Coverage of the nominal 95% interval $\hat{x}_j \pm 1.96\, s_j$ per parameter (points) and its mean (bar). (D) Twice the cost decrease from the truth to the estimate against χ^2_{35} quantiles. All fits use LSODA at relative tolerance 10^{-11} (Methods).

$|\mathcal{N}(0,1)|$; values below it in the high-WNT regimes are the non-asymptotic shrinkage of Section 2.4), whereas the integral PINN's errors are 13 to 35 times the bound at the median over both datasets (Fig 5B,C). The shortfall is not concentrated in the poorly informed parameters: parameters that the design pins to 10^{-3} are recovered by the PINN to 10^{-1}. This is an estimator gap in the plainest sense: the information is in the data, a classical estimator extracts it, and the PINN does not.

The classical fit needs care too. From the common starting guess, the relative-weighted fit reaches the global minimum in every regime, but in Early Adenoma only after 536 objective evaluations, beyond the cap of 400 used elsewhere; the unit-weight fit from the same guess converges to a genuine local minimum (restarting from it with stricter stopping moves it by 10^{-5}) at a cost 18 times that of the global one (19 of 34 recovered). Of the 36 jittered starts, 34 reach the global minimum (two in Early Adenoma stop in a local one at 27 of 34), and selecting among four starts by cost, which needs no knowledge of the truth, recovers the bound's count on both datasets (on the replicate 34, 34, 33 and 31, or 132 against a CRLB-expected 132.5). A few starts are cheap insurance against a likelihood with more than one basin.

2.6 Closing the estimator gap: a Schur-complement Gauss–Newton PINN

Where does the PINN lose the information? Not in the representation, which reaches a few percent (Section 2.2), and not in the objective, whose large-λ limit is the classical one; in the optimisation. A first-order method on a loss whose curvature spans six to nine orders of magnitude (Fig 3A) moves the well-informed directions quickly and the rest barely,

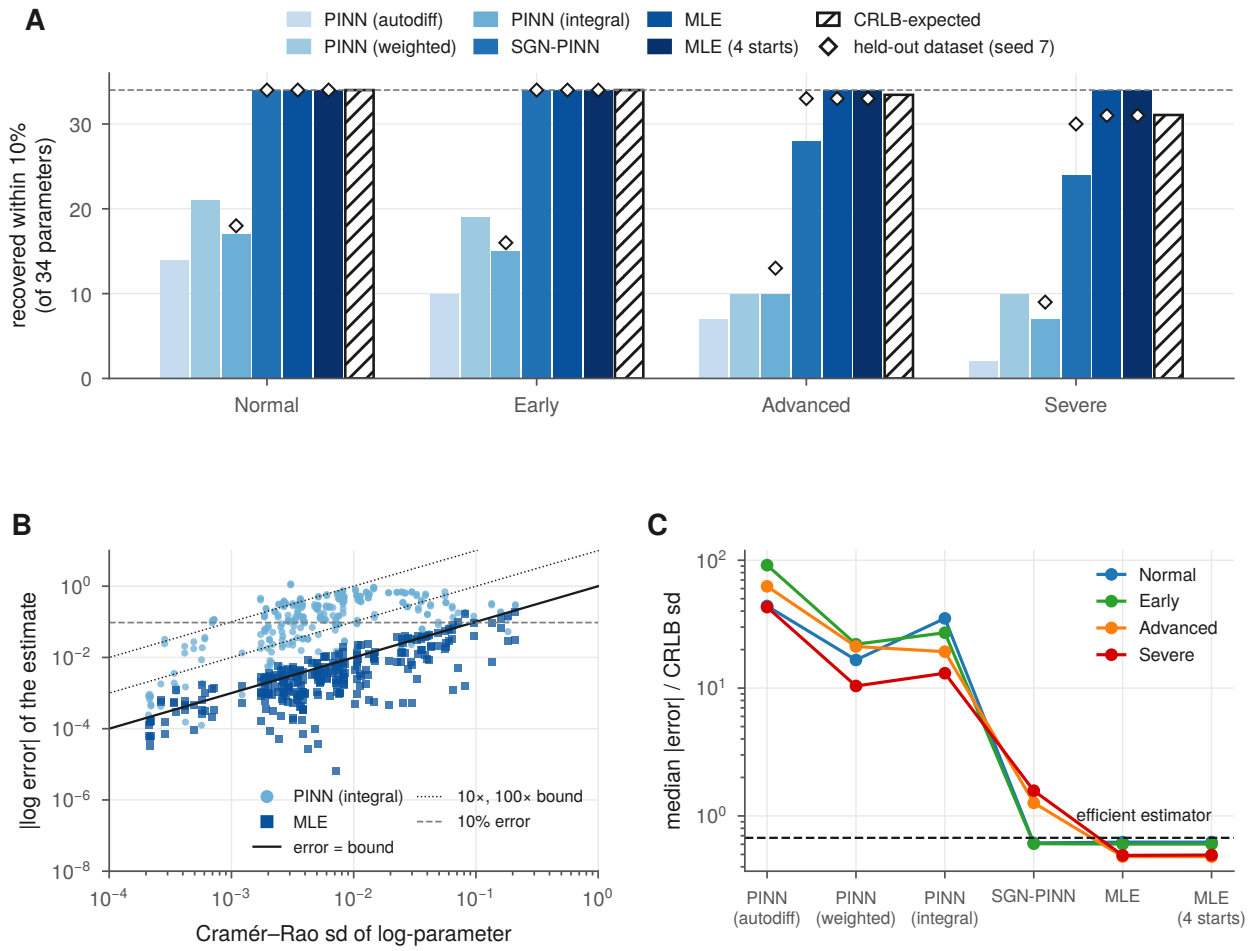


Fig 5. The estimator gap. (A) Parameters recovered within 10% (of the 34 outside the degenerate (θ_P, δ_{P1}) pair) per regime, for PINNs with the automatic-differentiation residual, with relative weights and fixed collocation, and with the integral residual (one start each), the SGN-PINN (Section 2.6), the converged classical fit from the common start and the best of four starts; bars, benchmark data (seed 42); diamonds, replicate dataset (seed 7; the autodiff and weighted PINN variants were run on the benchmark data only); hatched, the CRLB-expected count. (B) Absolute log-error of every regular parameter against its CRLB standard deviation for the integral PINN and the classical fit, both datasets; solid line, error equal to the bound; dotted, 10 and 100 times the bound; dashed, 10% error. (C) Median ratio of error to bound along the estimator ladder, per regime (both datasets where available); dashed, an efficient estimator.

and the network absorbs what the parameters do not explain. The SGN-PINN (Methods, Eqs (11)–(13)) freezes the trained PINN's hidden features, adds fixed random Fourier features and decaying exponentials, and optimises only the linear output layer of each condition together with θ. The inverse problem then becomes a separable nonlinear least-squares problem whose Gauss–Newton matrix is block-arrowhead (Fig 6A), so a full Levenberg–Marquardt step in all 5.6×10^4 unknowns costs ten block factorisations and a 36×36 solve. The physics weight λ is continued upward from 10^{-2}, where the heads mostly fit the data, to 10^3, where the trajectories must solve the ODE. Two schedules are run from the integral PINN's estimate, and the reported one is chosen by the shooting misfit of the ODE solution at its parameters, which needs no knowledge of the truth. The feature basis and the schedules were chosen on the benchmark dataset; the replicate dataset is held out.

On the held-out dataset the SGN-PINN recovers 34, 34, 33 and 30 of the 34 regular parameters, with median errors of 0.11%, 0.20%, 1.1% and 1.5% (Fig 5A). In Normal, Early and Advanced the continuation converges: the count rises within the first few iterations at small λ and holds as the physics residual is driven down by three orders of magnitude (Fig 6B), and parameter by parameter its errors coincide with those of the classical fit on the same data (rank correlation of absolute log-errors 0.99–1.00; Fig 6C). In Severe APC Loss the physics residual stalls at 1.3×10^4, the trajectory is

not an ODE solution, and although 30 of 34 parameters are within 10% their errors no longer track the classical fit's. On the benchmark dataset the method converges in Normal and Early (34 and 34) and stalls in the two high-WNT regimes (28 and 24; residuals 7×10^3 and 1.5×10^4). The SGN-PINN therefore reaches the classical estimate from inside the PINN framework when its continuation converges, and the physics residual, which needs no truth, says whether it has. In the cases where one schedule converged to a trajectory far from an ODE solution (physics residual above 10^5), the shooting misfit was 0.5–19% higher and the truth-free selection discarded it.

The SGN-PINN also returns, as a by-product of its last step, the inverse of the Schur complement. Its square root tracks the CRLB of the design over three orders of magnitude, at 1.2 to 1.6 times the bound at the median across regimes and datasets (Fig 6D); the excess is expected, because the relative-weighted objective is itself less efficient than the bound (standard deviations a median 7–9% and at most 47% above it; Methods) and the Gauss–Newton curvature at finite λ is not exactly the Fisher information. In an unconverged fit the estimate can be badly wrong (one parameter in the benchmark Severe fit is reported 2500 times too precise), so the self-reported precision is only as good as the convergence check that accompanies it. The method costs 15–27 minutes per regime (both schedules) on one GPU on top of the PINN it starts from, against 40–43 minutes for that PINN, 2–32 minutes for one start of the classical fit and under four minutes for the bound itself (Fig 6E; all measured on a shared node under load). Its contribution is not speed: it shows that the estimator gap of a PINN can be closed without abandoning the network representation of the trajectory, which is what one needs when part of the dynamics is itself learned, as in the hybrid models below.

2.7 An information gap: APC functionality on the β-catenin plateau

Setting the estimator gap aside, what remains is what the bound itself withholds. In Severe APC Loss four parameters have CRLB standard deviations above 15% (Fig 3C), and the widest belongs to the parameter that defines the disease stage. With δ_{P1} known, so that the structural degeneracy is removed and nothing else, the bound on θ_P widens from 0.04% in Normal to 0.23%, 7.2% and 57% across the regimes, a factor of 1600. Part of the trend is a coordinate effect: the quantity the APC equation actually contains is the excess degradation $\kappa_P = \delta_{P1}(1-\theta_P)$, whose relative precision is $\theta_P/(1-\theta_P)$ times that of θ_P, and its bound widens from 0.69% (Early) to 7.2% and 19%. The trend survives the change of coordinates at a third of the magnitude, and it has a mechanism. As WNT drive rises, APC falls (time average over $t \geq 10$ 0.48 in Normal, 0.076 in Severe) and the APC-dependent removal of β-catenin, $\lambda_P pb$, falls from 33% to 9% of total β-catenin turnover; the observables then respond only weakly to the APC branch, whose only free knob is θ_P, and the log-sensitivity of β-catenin to θ_P falls 24-fold (Fig 7C). Since this is information the data lack, the estimators of the previous sections should not recover θ_P in Severe APC Loss to better than the bound, and they do not: across 24 independent datasets the converged maximum-likelihood estimate of θ_P scatters by 33% (against a bound of 57%, the non-asymptotic shrinkage of Section 2.4), far too much to stage the disease from data of this kind.

The bound also says what would close the gap, and what would not. Removing one base condition at a time (S1 Fig) shows that the sustained WNT-inhibition step carries most of the design's information on θ_P: dropping it inflates the bound 2.8-, 6.3- and 2.4-fold in the first three regimes, while dropping any one of the other nine conditions changes it by at most 17%. It matters least in Severe APC Loss ($\times 1.25$), because a step of -0.7 against a drive of 2.0 does not lower β-catenin much. Adding one condition that is informative in general but does not lower β-catenin, a further ATRA protocol or a second WNT pulse, narrows the bound by 2–4.3%. Adding one WNT knockdown narrows it in proportion to how far the knockdown lowers β-catenin: by a factor of 4 at $k_W = 0.5$, 22 at $k_W = 0.1$ and 33 at $k_W = 0$ in Severe APC Loss (Fig 7A; these design comparisons use 200 uniformly spaced observations per condition, so their absolute values differ slightly from the benchmark design's). With the two-condition pairs on the 150-point benchmark design, the bound on θ_P in Severe APC Loss is 57% under the base design, 53% with the two information-matched conditions and 1.8% with the depletion pair, and the number of parameters with a bound below 5% goes from 25 to 25 and to all 35, respectively; in Advanced Adenoma the bound goes from 7.2% to 6.9% and to 0.42%.

These are local bounds, so the comparison was repeated by fit, with the converged maximum-likelihood estimator (here with unit weights, 36 free parameters, four starts, the best kept by cost) under the base design, the base design plus the two information-matched conditions, and the base design plus the depletion pair (a WNT knockdown $k_W = 0.02$ and a retinoid-free medium). Because θ_P alone is tied to δ_{P1}, the scored quantity is the identifiable product κ_P (Fig 7B). On this one dataset the fitted errors in κ_P are 0.35%, 2.2% and 5.1% (Early, Advanced, Severe) under the base design, each inside its bound (0.69%, 7.2% and 19%); 0.23%, 2.6% and 2.7% with the information-matched pair, whose bounds (0.64%, 6.9% and 18%) are barely narrower; and 0.20%, 0.29% and 0.25% with the depletion pair, whose bounds fall to 0.35%, 0.42% and 0.61%. A single dataset is one draw from the distribution the bound describes, and here the draws fall inside it in every cell; it is the bound, not the draw, that says the depletion pair is worth a factor of 17–31 in the high-WNT regimes and the information-matched pair almost nothing. The selection among four starts mattered in three cells: in

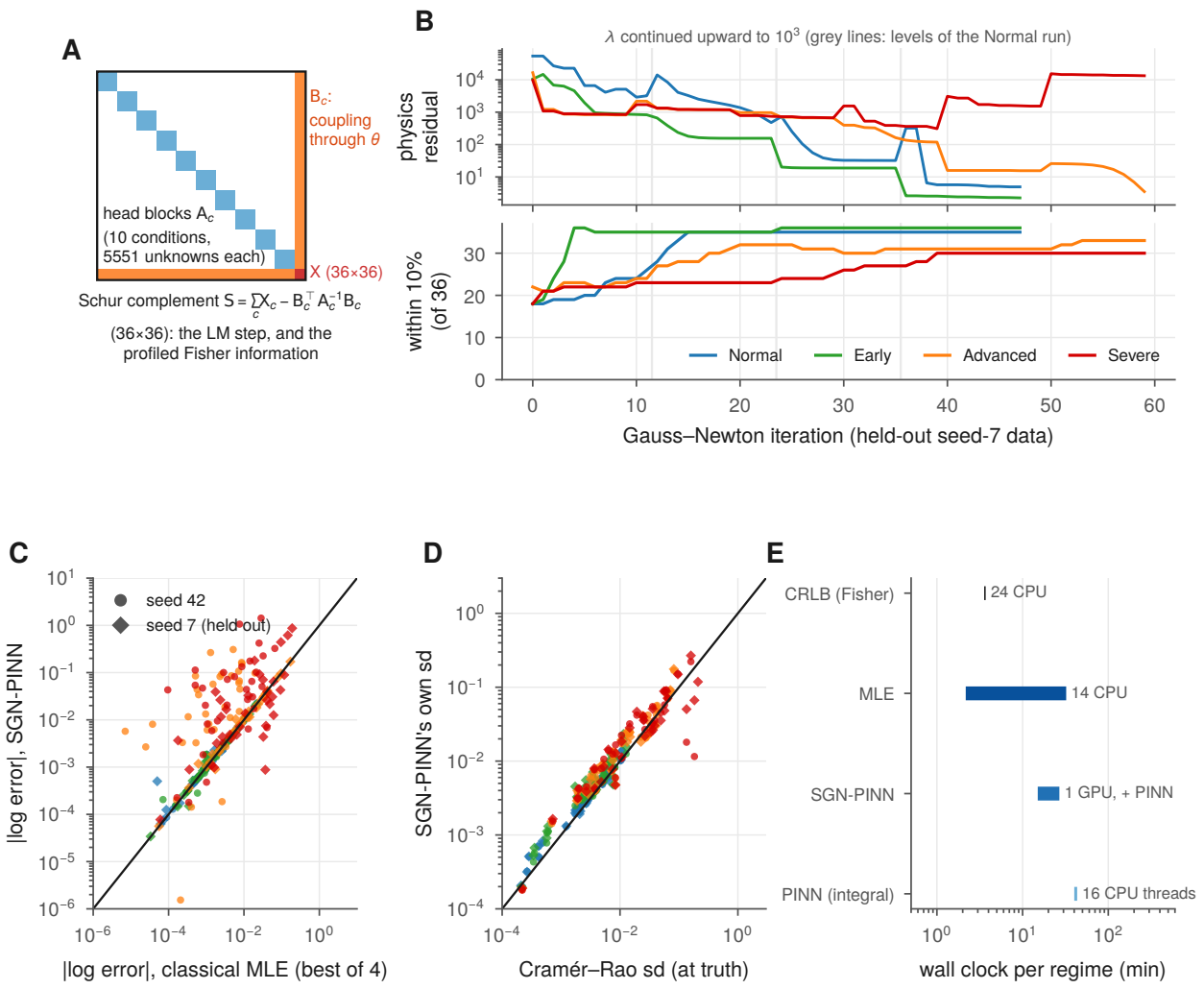


Fig 6. The Schur-complement Gauss–Newton PINN. (A) Structure of its Gauss–Newton matrix: dense blocks for the network heads of the ten conditions, coupled only through the 36 parameters, so each step reduces to the 36×36 Schur complement, which is also the profiled curvature that yields standard errors. (B) Physics residual (top) and parameters within 10% of 36 (bottom) along the continuation in λ (vertical lines) on the held-out dataset, per regime, for the schedule chosen by shooting misfit. (C) Absolute log-error of every regular parameter, SGN-PINN against the classical fit on the same data. (D) The SGN-PINN's own standard errors against the CRLB of the design. (E) Measured wall-clock time per regime (range over regimes and datasets; the node was shared, so times are indicative).

Early Adenoma the common starting guess converges to a local minimum under the base and information-matched designs, and in Normal under the depletion design it stops at the evaluation cap far from the minimum; in each case the cost picks the global one.

2.8 A learned regulatory term opens an information gap of its own

A neural–mechanistic hybrid replaces one closed-form regulation with a small network and keeps the rest of the model; for the β-catenin→MYC induction, $\eta_{BM} H_2(b; \kappa_{BM})$ becomes a network $f_\phi(b)$ beside the basal production rate a_M (Eq (14)). Run through the full PINN pipeline under the ten base conditions, the hybrid learns the function well (normalised RMSE 6–10% in the four regimes) and loses the basal rate beside it: a_M is 32.3% and 52.1% off in the two low-WNT regimes, where the mechanistic model recovers it to 0.4% and 4.6% (Fig 8A; run `20260727_012448_bm_myc`). The RA→HOXA5 hybrid misses a_5 by 9–37% (`20260726_214316_ra_h5`), but there the mechanistic PINN misses it by more (98–123%), so the full pipeline cannot attribute that loss; the equation-local analysis below, with exact states, can. Is this an estimator gap or an information gap?

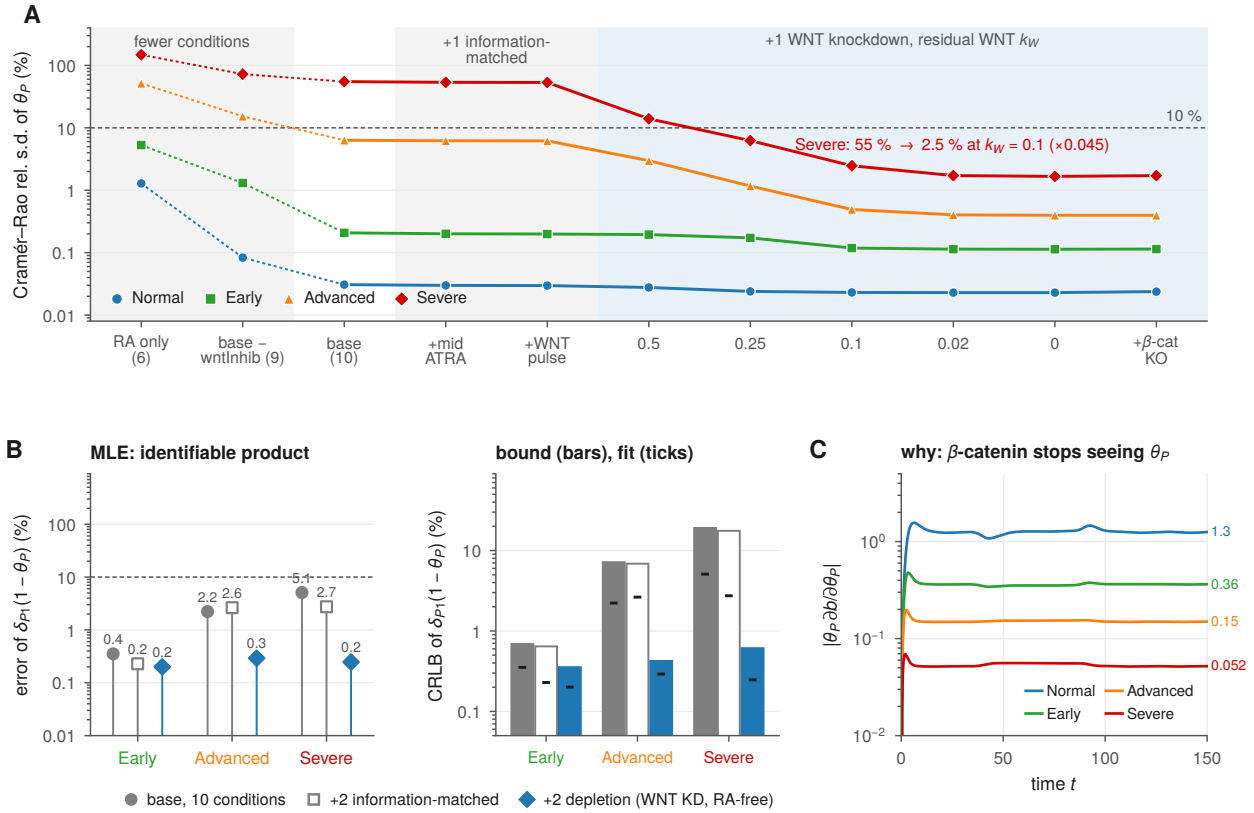


Fig 7. APC functionality is an information gap, and a WNT knockdown closes it. (A) Cramér–Rao relative standard deviation of θ_P with δ_{P1} known and 34 other parameters free, per regime, for designs ordered from fewer conditions (retinoid-only; base without the WNT-inhibition step), through the base design and the base design plus one information-matched condition, to the base design plus one WNT knockdown of residual strength k_W (and a β-catenin double knockout); 200 uniformly spaced observations per condition. (B) Converged maximum-likelihood fits of all 36 parameters (best of four starts by cost) under the base design, with two added information-matched conditions and with two added depletion conditions (WNT knockdown $k_W = 0.02$, retinoid-free medium), benchmark data: error in the identifiable product $\delta_{P1}(1-\theta_P)$ (left; zero by construction in Normal), and its Cramér–Rao bound on the 150-point benchmark design for each design (right, bars) with the fitted errors (ticks). (C) Log-sensitivity of β-catenin to θ_P under the control protocol: it falls 24-fold from Normal to Severe APC Loss.

The arithmetic suggests the latter. On the observed support, for any constant c,

$$a + f_\phi(u) = (a+c) + \big[f_\phi(u) - c\big], \tag{1}$$

and the bracket is a function the network represents as well as f_ϕ wherever the regulator u is bounded away from zero. A constraint $f(0) = 0$ (the *anchor*) removes the constant from the model at one point; it removes it from the fit only if the data visit that point. In the base design they do not. The lowest β-catenin level across all conditions and regimes is 0.036, an anchor ratio (Eq (15)) of 0.025–0.081; in the two high-WNT regimes the lowest value, 0.20, is the initial condition itself, so what little support near the anchor exists there is set by the choice of initial state, not by the dynamics. The lowest RA level is 0.136. In the Early Adenoma fit the learned MYC activation satisfies $f_\phi(0) = 0$ exactly, but over the observed range ($b \geq 0.11$) it sits a near-constant 0.094 above the true Hill function, and a_M is low by the same 0.094 ($\hat{a}_M = 0.086$ against 0.18; Fig 8A). The identity is close to automatic for any fit that reproduces the production flux with the correct timescale; what is informative is that the offset is nearly constant across the support. Across the sixteen equation-local fits of four network parameterisations in four regimes (below) the mean offset equals the basal deficit to within 0.005 (Fig 8B). In the full pipeline it holds in Early Adenoma and approximately in Normal (offset 0.067 against a deficit of 0.058), and not in the two high-WNT regimes; there, and in Normal, the fit also inflates the MYC timescale ε_M (0.69–0.72 against 0.60). This is the constant-offset form of the degeneracy that Loman and Baker construct for a linear-rate term [26], and it is consistent with an instance reported in the literature, a learned nuclear-export term recovered by symbolic regression as $3.96 \times 10^{-4} x + 0.8$ against a true $6.17 \times 10^{-2} x$ [11].

If the gap were an estimator gap, a better-shaped network would close it. Loman and Baker found that monotone, bounded networks bring parametric identifiability on a par with the fully known model in a linear-pathway model [26]. To test this at scale we used an equation-local screen (Methods): only the equation that hosts the learned edge is fitted, from exact states, by the network and, from the same seeds and budget, by the mechanistic control. Because the states are exact, the screen has no measurement noise; its errors are those of the optimisation and of the model class, and its absolute values bound the full pipeline from below. Across the seven production edges of the model (S2 Table), a monotone network with an exact anchor (`sc`) sometimes halves the basal error of the non-monotone gated network (`gated`) and sometimes quadruples it; averaged over regimes, the gated network leaves the basal parameter 16–50% off and the monotone one 14–74%, against 7–25% for the mechanistic control (Fig 8C). In the two low-WNT regimes, where the control recovers all fourteen basal parameters to below 7.5%, the two networks recover 4 of 28. In the two high-WNT regimes the control degrades as well (all 14 of its failures are there), and the comparison is between two failures. For the MYC edge all five parameterisations were run in all regimes (Fig 8D): where the control recovers a_M to 0.0–0.1%, the gated, monotone and linear-anchor networks leave it 6.5–38% off, the monotone-rate form 200–400% off (it fails through the timescale, which it inflates to 1.2–18.3), and the bounded form 10.5–17.9%, although it is handed the true asymptote; in the two high-WNT regimes the bounded form (8.4%, 18.5%) beats a control that has itself degraded (15.6%, 35.6%). No network parameterisation closes the gap where the control is sound. Restart noise is large away from the anchor: the same MYC-edge cell (Advanced Adenoma, gated) reads 68.3% in the two-restart atlas and 18.9% in the three-restart five-form run, so cells are compared within an experiment, not across. Modulator edges, where the network *multiplies* a state ($f(u)\cdot y$) and there is no additive constant to absorb, behave differently: CYP26A1⊣RA and HOXA5⊣β-catenin are learned to 0.0–1.7% functional error with every surviving equation parameter recovered in the low-WNT regimes (run `20260801_204046_screen_atlas`).

2.9 The Fisher information of a learned term

The screen has no measurement noise, so its Fisher information is formally infinite. To ask the same question of it that Fig 3 asked of the mechanistic model, we treat the residual of the hosting equation as noisy with an arbitrary scale σ_r and compute Γ, the CRLB of the basal parameter per unit σ_r, with the learned term represented by the screen's own network, fitted to the true regulation and profiled out (Methods). Γ is then a measure of how well the design conditions the basal parameter against everything the network can do; we report it relative to the same quantity for the mechanistic control, $\Gamma_{\text{net}}/\Gamma_{\text{mech}}$, the information lost to the network. For a network that is not observed near its anchor the network block is nearly rank-deficient, and the absolute value of Γ depends on the numerical rank assigned to it, by a factor of 17 to 5000 between relative tolerances 10^{-6} and 10^{-10}; the size of the base-design loss is therefore not determined, while its ordering, the design factors where the anchor is visited, and every rank correlation below (which move by at most 0.04) are. We quote values at tolerance 10^{-8}. Against Monte Carlo refits of the MYC edge it is calibrated where the anchor is visited (empirical and predicted standard deviation within 3% for the network and 1% for an anchored cubic). Where it is not visited, the bound at tolerance 10^{-8} overstates the empirical scatter of a Levenberg–Marquardt refit 23-fold, and a tolerance of 10^{-5} brings it to within 12%; the size of the base-design loss quoted below is therefore an upper estimate, and only orderings and ratios are used in what follows.

Three results (Fig 9). First, under the base design the network loses a factor of 56 to 4×10^5 in basal-rate precision relative to the mechanistic control of the same equation, and the loss is a property of the network's flexibility on the unobserved interval: an anchored cubic, which cannot bend freely between $u=0$ and $\min u$, is within a factor of four of the control. Second, two information-matched conditions change Γ by a factor 0.90–1.00 for the network and the mechanistic control alike (0.64–0.99 across all representations), whereas a depletion pair that reaches the anchor of the RA-driven edges shrinks it by a factor of 1.8–6.1×10^{-4} and brings the network to 1.0–1.4 times its control (Fig 9A); for the mechanistic control the same pair helps three- to twentyfold. Depletion does not add general information, it removes the loss that the network introduced. For the β-catenin→MYC edge, the double knockdown that takes β-catenin to zero brings the network to its control in every regime, while the single WNT knockdown, which leaves β-catenin at 1.3–1.5% of its range, leaves a loss of 65–210. Within each dose arm, Γ falls monotonically with the achieved anchor ratio (Spearman ≥ 0.98 in all six arm–regime series, Fig 9B), but the arms do not collapse onto one curve: at matched $\rho\approx0.016$ in Severe APC Loss the loss is 3.5 for the retinoid arm and 230 for the β-catenin arm. The anchor ratio orders designs within an arm; the bound weighs them across arms.

Third, the bound is consistent with the fits, which were all run before it was computed. Over 116 distinct screen fits (atlas, depletion and information-matched comparisons, dose-response, prospective test), the rank correlation between Γ and the measured basal error is $+0.79$, against $+0.55$ for the anchor ratio alone (Fig 9C). The correlation is carried by design changes: within the base design, where edges differ in equations rather than in what the data visit, it is

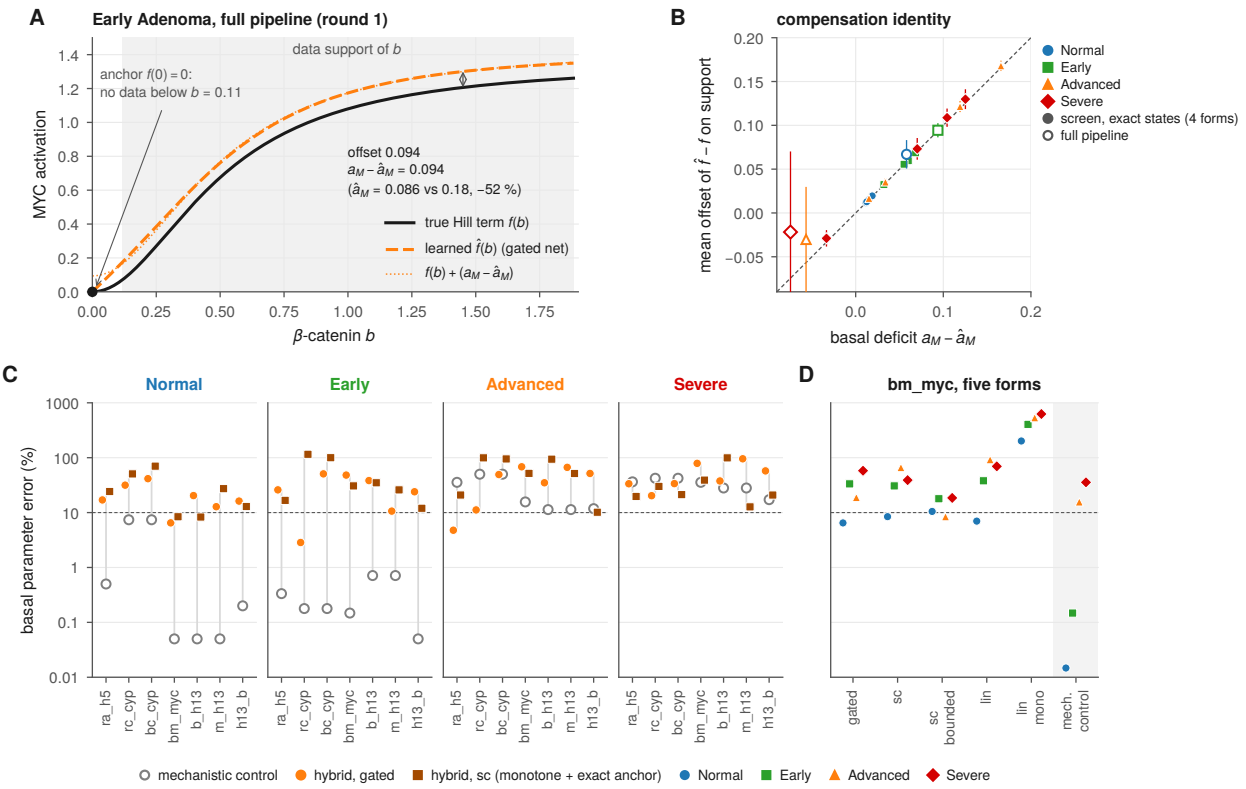


Fig 8. A learned term loses its host's basal rate. (A) Full-pipeline β-catenin→MYC hybrid, Early Adenoma, ten base conditions: the learned activation (dashed) obeys $f_\phi(0) = 0$ but no data lie below $b = 0.11$ (grey band, data support); over the support it sits 0.094 above the true Hill term (solid), and the basal rate a_M is low by the same amount; dotted, the true term shifted by the basal deficit. (B) Compensation identity, Eq (1): mean offset of the learned function over the support against the basal deficit, for the equation-local fits of four network parameterisations (filled) and the full pipeline (open, which departs from the identity in the two high-WNT regimes, where the fit also inflates the MYC timescale); dashed, $y = x$. (C) Basal-parameter error of the seven production edges under the gated and the monotone (`sc`) network and the mechanistic control (open), ten base conditions, two restarts; Normal values are from the run log at 0.1% precision and a reported 0.0% is drawn at 0.05%. (D) Basal error of a_M under all five network parameterisations and the mechanistic control (three restarts; a separate run from C).

$+0.32$ ($n = 56$); and the mechanistic Γ correlates with the controls' errors about as well ($+0.80$), so Γ measures the conditioning of the equation as much as the network's loss. The cells share edges and regimes, so we give no p-value. One discordance is instructive. Under the single-knockdown depletion pair the bound predicts a similar residual loss for the β-catenin→MYC and β-catenin→CYP26A1 edges (65–210 and 23–142), yet the fit improved the former thirty-fold and the latter not at all (below): the bound ranks designs, not individual fits.

Two edges show what the bound adds to the anchor ratio. For MYC→HOXA13, neither the depletion pairs nor a β-catenin double knockout lowers the loss below 2.6×10^3 in any regime, although the latter cuts the MYC level; only a protocol that also silences basal MYC transcription (`mycKO`: the β-catenin double knockout plus $k_{aM} = 0$) brings it to 1.01 in all four regimes. For HOXA13→β-catenin, whose basal parameter is W itself, every protocol that drives HOXA13 to zero leaves the loss as large as it was or larger (6.5×10^4 to 7.7×10^5): HOXA13 can only be silenced by first removing β-catenin, whose production is $k_W W + k_{13}\eta_{13}H_2(h_{13})$, so the conditions that visit the anchor set $k_W = 0$ and delete W from their own equations. The bound is consistent with the second clause of the design rule that the prospective test below arrived at empirically. It also points to a remedy the rule missed: the *partial* WNT knockdown $k_W = 0.02$, which does not reach the anchor ($\rho = 0.16$), brings the loss for W to 1.6–2.0, because two conditions that observe W and $0.02\,W$ with a known knockdown efficiency identify W with no anchor at all.

2.10 Designs that visit the anchor restore the basal rate

The fits that the bound summarises are, in order of stringency, as follows. Adding the depletion pair (the WNT knockdown $k_W = 0.02$ and a retinoid-free medium, which drives RA to zero exactly) to the ten base conditions takes the basal error

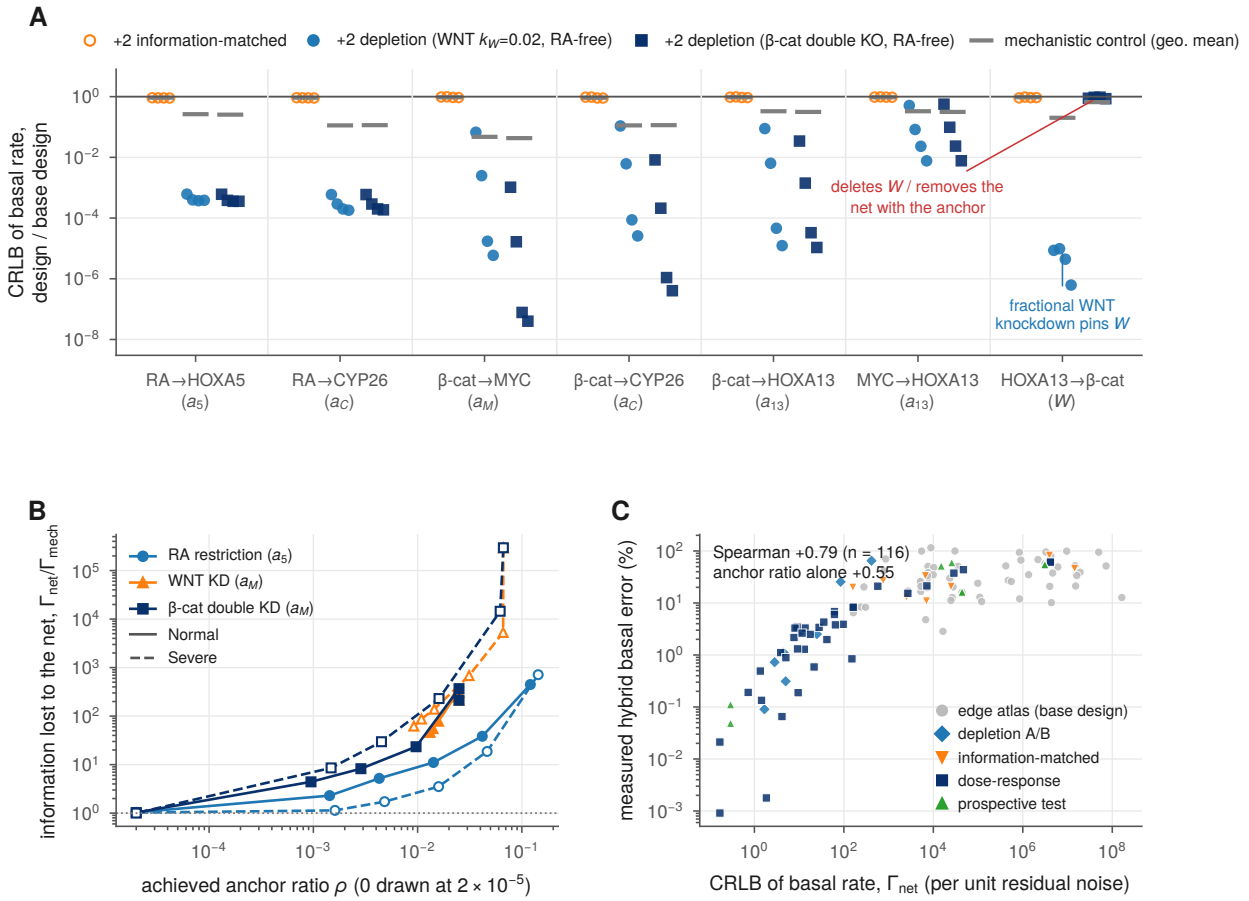


Fig 9. The Fisher information of a learned term. Cramér–Rao bound of the basal parameter of each production edge in the equation-local problem, per unit residual noise, with the learned term represented by the screen's gated network and profiled out; training-free. (A) Bound under an added design relative to the base design, per regime (points): two information-matched conditions (open), two depletion conditions with the single WNT knockdown (circles) or the β-catenin double knockout (squares); grey bars, the same factor for the mechanistic control. (B) Information lost to the network, $\Gamma_{net}/\Gamma_{mech}$, along the three dose arms of Fig 11 (solid, Normal; dashed and open, Severe APC Loss; ratio zero drawn at 2×10^{-5}). (C) Bound against the measured basal error of 116 distinct screen fits, coloured by experiment.

of the two RA-driven edges from 10.9–33.5% to 0.09–1.1%, and the RA→HOXA5 edge in Severe APC Loss from no equation parameter recovered to all four (Fig 10A). The MYC edge, whose regulator only approaches the anchor, improves up to thirty-fold but stops at 2.5–2.9%; the β-catenin→CYP26A1 edge does not improve (24.6%→25.5% in Normal, 30.0%→64.8% in Severe). Adding instead the information-matched pair, two conditions of the same size and style that leave every anchor ratio unchanged, changes the basal error by a factor of 0.76–1.54 in both directions and the equation parameters recovered from 16 to 15 of 28, whereas the depletion pair cuts it 7- to 200-fold on three of the four edges (not on β-catenin→CYP26A1) and recovers 26 of 28 (Fig 10B). These two comparisons used one restart, and away from the anchor single cells move by tens of points between restarts; the replicated contrast is the next one.

At fixed experimental effort. Depletion and information-matched conditions differ in what they reach and also in how strongly they perturb. A dose-response with the number of conditions fixed separates the two (Fig 11A,B). At every dose the design is the ten base conditions plus one graded depletion of depth $k \in \{1, 0.3, 0.1, 0.03, 0.01, 0\}$, so no dose has more data than another and $k = 1$ is each arm's own null. The three arms are graded retinoid restriction (scoring a_5); a graded WNT knockdown alone (scoring a_M), under which β-catenin plateaus at about 1% of its range because the HOXA13 feedback keeps producing it; and the same knockdown applied to both sources of β-catenin ($k_W = k_{13} = k$), which reaches zero. The last two score the same edge, equation and parameter with the same seeds, budget and number of conditions, and at $k = 0$ differ by one knockdown. Replicated with five restarts from three disjoint seed sets, the

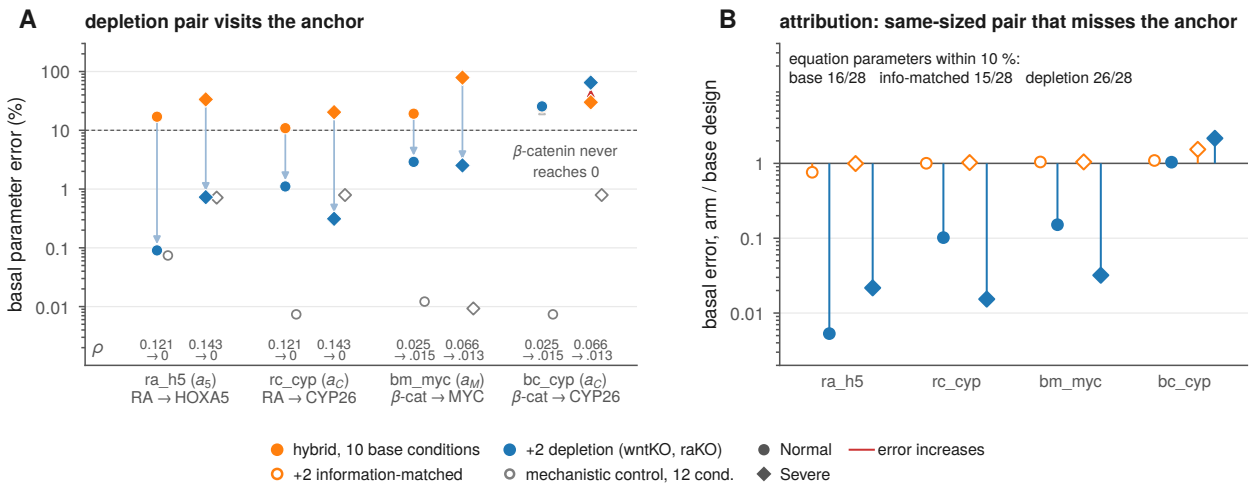


Fig 10. Visiting the anchor, not adding data, restores the basal rate. Equation-local screen, gated network, one restart. (A) Basal error under the ten base conditions (orange) and with two depletion conditions added (blue), for four production edges in Normal (circles) and Severe APC Loss (diamonds); grey, mechanistic control on the same twelve conditions; anchor ratios before and after below each edge. (B) Ratio of the basal error to its base-design value with two information-matched conditions (open orange) and with two depletion conditions (blue).

single knockdown leaves a_M at 1.45% (range 1.33–1.52) in Normal and 2.25% (1.79–2.63) in Severe APC Loss; the double knockdown at 0.003% (0.001–0.008) and 0.002% (0.001–0.003), 430- and 1200-fold lower (ratio of the means), against a seed spread of 0.2 and 0.8 points. The single knockdown at $k = 0$ removes the WNT drive completely, and the mechanistic control is then at 0.0%; β-catenin, still produced through the HOXA13 feedback, stops at about 1% of its range, and the network's basal error stops with it. Against dose the arms separate. Against the achieved anchor ratio the two β-catenin arms line up (at matched ratio they agree within a factor of about two) and the single-knockdown arm sits at its ratio floor, but the retinoid arm does not join them in Severe APC Loss: there it stays near 3% down to ratio zero, at parity with its mechanistic control (3.4%), whose own information about a_5 is limited in that regime. The anchor ratio orders doses within an arm; how much a given ratio is worth differs between edges, as the Fisher analysis predicted (Fig 9B). From its own null dose, the MYC edge in Severe APC Loss goes from 61% to 0.001%. The monotone exact-anchor network behaves the same way (retinoid arm 22.8%→0.0% in Normal; double knockdown 70.3%→0.0% in Severe; single knockdown stops at 1.8% and 2.4%).

Six predictions for this experiment were registered before it was run (S1 Text). As registered, with two restarts in Severe APC Loss, two of the six passed in both regimes; rerunning Severe with five restarts, which removed one bad optimum, three pass. The failures include the registered falsifier in Normal: we had predicted that the single knockdown would never fall below 2%, and it reached 1.3%; at matched anchor ratio the two β-catenin arms agree to within a factor of 2.0 (Normal) and 2.3 (Severe), which is what the anchor account predicts, while at ratio zero the double knockdown is three orders of magnitude lower.

With state estimation in the loop, the full pipeline confirms the contrast. With the double knockdown and the retinoid-free medium added, the MYC basal error falls from 10.3/11.7/17.4/16.4% under the single knockdown to 1.4/0.6/0.6/0.7% across the regimes, against a mechanistic control of 0.2–0.3% on the same twelve conditions; the hybrid recovers 58 of 136 shared regime–parameter pairs against the control's 62, three of the four lost pairs in the MYC equation (S3 Table). One further knockdown turns a 10–17% error into one within about a point of the fully mechanistic fit; the full-pipeline errors are larger than the screen's, as expected from state-estimation error.

A prospective test. Because the anchor ratio needs only reference trajectories, it can be tabulated for a ladder of candidate protocols ordered by laboratory cost before any fit (S4 Table), naming for each learned edge the cheapest protocol that puts its anchor in the data. The table was tested on the two production edges with the largest functional error in the atlas, MYC→HOXA13 and HOXA13→β-catenin, neither of which had been run under any depletion, each in three arms: the base design; a *near-miss*, a larger perturbation than any in the base design that leaves the edge's anchor ratio far from zero; and the prescribed protocol (Fig 11C). For MYC→HOXA13 the prescribed protocol (`mycKO`) took

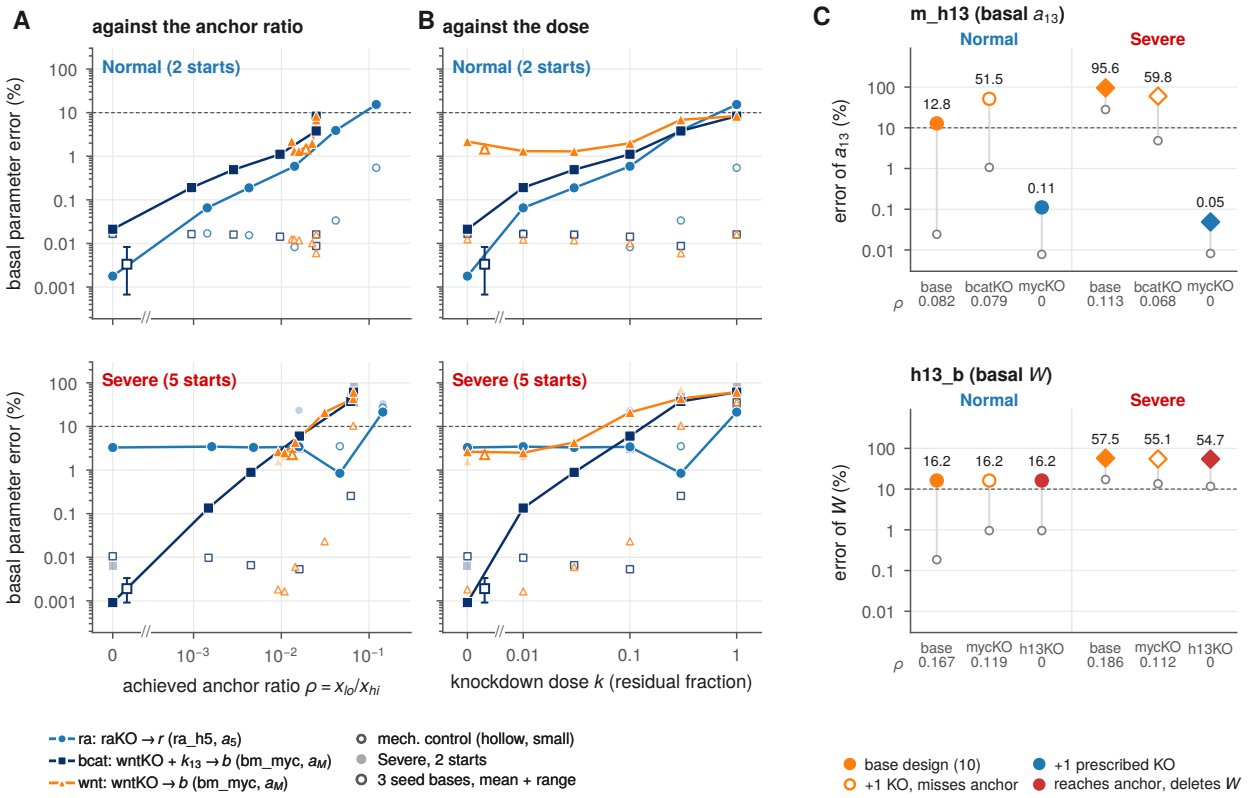


Fig 11. Reaching the anchor, not perturbing harder, restores the basal rate. (A, B) Dose-response with eleven conditions at every dose: network basal-parameter error (filled) and mechanistic control (open, small) against the achieved anchor ratio ρ (A) and against the knockdown dose k (B), for graded retinoid restriction (a_5), graded WNT knockdown (a_M) and graded knockdown of both β-catenin sources (a_M). Normal two restarts; Severe five restarts, two-restart values faint; squares with bars at $k = 0$, mean and range over three disjoint seed sets of five restarts. (C) Prospective test: basal error under the base design, under a near-miss that does not reach the anchor (open) and under the prescribed protocol, for MYC→HOXA13 (top, basal a_{13}) and HOXA13→β-catenin (bottom, basal W; the HOXA13 knockout, labelled h13KO, reaches the anchor but deletes W, red); grey, mechanistic control; anchor ratio below each arm.

the basal error from 12.8% to 0.1% in Normal and from 95.6% to 0.05% in Severe APC Loss, while the near-miss, the β-catenin double knockout, which moves the anchor ratio from 0.082 to 0.079 (Normal) and 0.113 to 0.068 (Severe), left it at 51.5% and 59.8%, in Normal worse than doing nothing. For HOXA13→β-catenin the registered prediction failed: the basal error did not move under any arm (16.2% in all three in Normal, 57.5% to 54.7% in Severe). The protocol that reaches its anchor, a HOXA13 knockout ($k_{a13} = 0$ on top of `mycKO`), necessarily includes $k_W = 0$ and so deletes the parameter it was meant to identify. The design rule therefore has two clauses, *reach the anchor, with a condition that still contains the basal term*; the second was formulated after this failure, and the Fisher analysis above is consistent with it. Of the five registered predictions, the two that carry the claim passed (the prescribed arm below 3%; the near-miss above 8%), and three failed (S1 Text). The prescribed protocol also made the *functional* error of the MYC→HOXA13 network worse (57%→68% in Normal, 12%→81% in Severe) while it fixed the parameter: enlarging the support towards zero exposes the network to inputs on which it was never constrained, so functional and parametric identifiability moved in opposite directions, the distinction Loman and Baker draw [26] made concrete.

3 Discussion

The accounting. The question this paper has asked at every level of the problem is whether a parameter that is not recovered was lost by the estimator or never present in the data, and the answer has come from one computation, the Fisher information of the experiment at the truth. The answer was different at different levels. For the 34 regularly behaved kinetic parameters of the mechanistic model, the bound said nearly everything is in the data, and the failures of earlier PINN designs, which had been read as a hard ceiling of the model, were an estimator gap: a converged classical

estimator reaches the bound, and a Gauss–Newton PINN reaches the classical estimator where its continuation converges. For APC functionality in the high-WNT regimes and for the basal rate beside a learned term, the bound said the data do not contain the parameter, and no estimator change we tried, whether a different residual, a second-order optimiser, a converged maximum-likelihood fit or five different constraint parameterisations of the learned term, recovered it. In both cases the bound also names the remedy: a condition that leaves the β-catenin plateau for θ_P, and a condition that visits the anchor while still containing the basal term for the learned production edges. Each remedy is confirmed by fits (for the learned term, fits run before the bound was computed), against an information-matched control that added the same amount of data without visiting the relevant region.

Why the bound is the right yardstick for PINNs. A PINN's parameter-recovery count is a property of the estimator and the data jointly, and without a yardstick it cannot be attributed to either. The Cramér–Rao bound costs one sensitivity computation (about 720 ODE solves per regime here, under four minutes on 24 cores). It is local and asymptotic, and both qualifications matter. We therefore verified that the bound is attained, by Monte Carlo over independent datasets (Section 2.4), and we reported where it is not exactly attained: in the two high-WNT regimes the converged estimates scatter less than the bound for the most weakly informed parameters. The likelihood also has more than one basin, and the classical estimator from a single start can be trapped (in Early Adenoma it was); a few starts selected by cost were enough here, but in general the gap between a fit and the bound can be partly an optimisation gap of the likelihood itself, not of the PINN. Profile likelihood [14, 27, 28] is the standard non-local alternative; it requires a full re-optimisation of the other 35 parameters at every point of every profile, and for the questions asked here (which parameters does this design inform, and which added condition informs them) a bound computed once and checked once by Monte Carlo, was sufficient. The practical recommendation is modest. On synthetic benchmarks, compute the bound of the design before training and report recovery against it: a parameter whose error exceeds the bound by orders of magnitude is an estimator problem. On real data the truth is unknown, but the same comparison can be made between estimators: the bound at the estimate predicts how far two estimators on the same data should agree, a fit whose parameters differ from a converged classical fit by many bound-widths has not converged, multistart costs reveal the multimodality that makes any local bound misleading in a wrong basin, and for the SGN-PINN the physics residual says whether its continuation reached an ODE solution.

The estimator side. The Schur-complement Gauss–Newton PINN is not a new optimiser; it is the observation that freezing a trained PINN's hidden features, so that each state is linear in its output layer, turns the inverse problem into a separable nonlinear least-squares problem, for which Gauss–Newton with variable projection is the classical tool [29–31], and that the block-arrowhead structure of its normal equations makes each step cost ten small factorisations. Random-feature and extreme-learning-machine PINNs [32–34] exploit the same linearity for forward problems; second-order and natural-gradient optimisers for PINNs [35, 36] are a related line. What the formulation adds for inverse problems is that the Schur complement it already computes is the head-profiled Gauss–Newton curvature, which approximates the profiled Fisher information when the physics term is satisfied, so the estimator reports an estimate of its own CRLB at no extra cost. That estimate of uncertainty is only as good as the point it is computed at: where the continuation stalled, the SGN-PINN reported confident standard deviations around wrong estimates, which is the general limitation of any local uncertainty estimate [37].

The information side, and its relation to prior work. The degeneracy behind the learned-term result is known: Loman and Baker construct it for a linear-rate term and note its generality [26]. They also find that monotone, bounded constraints bring parametric identifiability on par with the fully known model, which is not what our monotone exact-anchor or oracle-bounded forms achieve under the base design. We think the difference is the variable they do not treat, the data support of the learned function: the same forms recover the basal rate the moment one condition reaches the anchor. The survey of Philipps et al. [11] reports a learned term recovered with a spurious additive constant and attributes the mismatch to non-identifiabilities introduced by the network; the compensation identity, Eq (1), says precisely which constant, and the hybrid Fisher information says which experiment would have removed it. To our reading the only optimal-experimental-design work for universal differential equations [38] designs on the joint Fisher information of network weights and mechanistic parameters without shape constraints, so it has no notion of designing a condition so that a constraint binds; classical design for dynamic models [21–23, 39, 40] optimises a scalar of the Fisher matrix over inputs and sampling times. The anchor ratio is a cruder, cheaper and constraint-aware surrogate, and the hybrid Fisher information is its quantitative version. In the language of adaptive control, both information gaps are failures of

persistence of excitation [24, 25]: a learned map is identified only along the trajectory the system follows, and so, we add, is any constraint imposed on it.

Biological reading. The model is a reduced caricature of the crypt, but two of its conclusions translate into experimental advice that does not depend on its details. APC functionality, the quantity that distinguishes the disease stages, is the parameter that a perturbation design built around retinoid treatment and WNT/MYC pulses informs least exactly where it matters most, in the high-WNT states, because β-catenin saturates the readouts that would report it; a graded WNT knockdown restores it, while two added conditions that perturb without lowering β-catenin do not. And when a regulatory law is learned by a flexible network rather than assumed, the basal transcription rate of the target gene, a quantity of direct biological interest, is identified either by a condition that silences the regulator, which in a positive-feedback network may require knocking down the feedback as well as the input (the single WNT knockout leaves β-catenin at about 1% of its range because HOXA13 keeps producing it), or by a known partial knockdown of the basal term itself.

Limitations. Every result is on synthetic data generated by the model itself, with all seven states observed at 150 time points with noise of 0.1–2.6% of their mean levels, so every recovery count is an upper bound on what real measurements of a subset of states would give, and the regimes are not calibrated to patient data. The information budget depends strongly on that design (S1 Fig): with only β-catenin, the two HOX genes and MYC observed, an efficient estimator would recover 16–30 of the 34 regular parameters per regime, with β-catenin and MYC alone 4–8.5; with ten-fold larger noise and all states observed, 17–32. The accounting itself does not depend on the design, and it is exactly what would tell a modeller which of these regimes they are in before any network is trained. The Cramér–Rao analysis is local and asymptotic: in the two high-WNT regimes it is conservative for the weakest parameters, and the likelihood has more than one basin. The hybrid experiments other than the full-pipeline confirmations use an equation-local screen with exact states; the full pipeline was rerun for two edges and confirms the direction of the effects, with larger absolute errors than the screen's because state estimation is in the loop. Restart counts move individual hybrid cells by up to fifty points away from the anchor, so numbers are compared within experiments; the headline contrast at the anchor is replicated over three disjoint seed sets. As registered, two of the six dose-response predictions passed (three after rerunning one regime with more restarts) and two of the five prospective-test predictions (S1 Text), including the registered falsifier failing in one regime; the second clause of the design rule was formulated after the failure that revealed it, and the hybrid Fisher information reproduces it but was computed afterwards. The prescribed protocols for the MYC and HOXA13 edges require silencing a constitutive promoter, which is materially harder than the medium change that fixes the RA edges, and the Fisher analysis points to a cheaper alternative it did not test by fit: a known partial knockdown of a basal term identifies it without visiting the anchor. Every knockdown here, and that alternative in particular, assumes the knockdown efficiency is known exactly; real siRNA efficiencies are uncertain, and an unknown efficiency adds one parameter per knockdown condition that a design would have to identify too. Finally, no symbolic structural-identifiability analysis [41, 42] was run; apart from the $\delta_{P1}(1-\theta_P)$ product, every identifiability statement here is practical.

Conclusion. A recovery count does not say whether to change the estimator or the experiment; the Cramér–Rao bound of the design does, at the cost of one sensitivity computation. In a seven-state model of colorectal-cancer signalling it showed that the long-standing recovery ceiling of physics-informed estimators was numerical, removable by a converged classical fit and by the SGN-PINN where its continuation converges, that the parameter defining the disease stage is uninformed by the conventional design in exactly the states that matter, and that the information a hybrid model's constraint carries is set by where the experiment takes the regulator. For neural and hybrid inverse problems in systems biology, the first diagnostic should be the information budget of the design.

4 Materials and methods

4.1 The WNT–RA–HOX model

The model is a reduced, nondimensional description of the antagonism between WNT/β-catenin signalling, which sustains proliferation and stemness in the colonic crypt, and retinoic acid signalling, which drives differentiation through the HOX transcription factors [3, 43–46]. The states are β-catenin b, APC p, HOXA5 h_5, HOXA13 h_{13}, MYC m, RA r, and the RA-catabolising enzyme CYP26A1 c. With $H_n(u;\kappa) = u^n/(\kappa^n + u^n)$ and Hill exponents $n = 2$ for every β-

catenin, MYC and HOXA13 input,

$$\dot{b} = W + \eta_{13} H_2(h_{13};\kappa_{13}) - b - \lambda_P\, p\, b - \lambda_5 \frac{h_5 b}{\kappa_5 + b}, \tag{2}$$

$$\varepsilon_P \dot{p} = \frac{1+\rho_5 h_5}{1+\rho_B b + \rho_{13} h_{13}} - \delta_P\, p, \qquad \delta_P = 1 + \delta_{P1}(1-\theta_P), \tag{3}$$

$$\varepsilon_5 \dot{h}_5 = a_5 + \eta_R H_1(r;\kappa_R) - h_5 - \eta_M \frac{m h_5}{\kappa_M + m}, \tag{4}$$

$$\varepsilon_{13} \dot{h}_{13} = a_{13} + \eta_{B13} H_2(b;\kappa_{B13}) + \eta_{M13} H_2(m;\kappa_{M13}) - h_{13}, \tag{5}$$

$$\varepsilon_M \dot{m} = a_M + \eta_{BM} H_2(b;\kappa_{BM}) - m, \tag{6}$$

$$\varepsilon_R \dot{r} = \mu_R(t) - r - \lambda_C\, c\, r, \tag{7}$$

$$\varepsilon_C \dot{c} = a_C + \eta_{RC} H_1(r;\kappa_{RC}) + \eta_{BC} H_2(b;\kappa_{BC}) - c. \tag{8}$$

Time is scaled by the β-catenin turnover rate and the horizon is $t \in [0, 150]$. The retinoid supply is a basal level, a circadian dietary term and a smooth all-*trans* retinoic acid (ATRA) pulse, $\mu_R(t) = \mu_0 + A_R[1 + \cos(2\pi t/T_R - \varphi)] + \frac{D_R}{2}[\tanh q(t-\tau_1) - \tanh q(t-\tau_2)]$, with $T_R = 24$ and a baseline treatment window $[\tau_1, \tau_2] = [40, 88]$. Exogenous WNT and MYC inputs, zero by default, are added to Eqs (2) and (6) as smooth steps of amplitude D_W and D_M when a condition calls for them, and knockdowns enter as known multiplicative efficiencies on the WNT drive (k_W), the HOXA13→β-catenin feedback (k_{13}) and the basal production terms ($k_{a5}, k_{a13}, k_{aM}, k_{aC}$), all equal to 1 unless stated. The initial state is $(0.20, 1.00, 0.80, 0.30, 0.30, 0.60, 0.40)$. The 36 kinetic parameters and their baseline values are listed in S1 Table; the forcing parameters are known. The stemness index plotted in Fig 1 is $S = b(1+h_{13})/(1+p+h_5)$.

Disease progression is represented by four regimes that differ only in WNT drive W and APC functionality θ_P (Table 1). APC loss enters through one scalar, the excess degradation δ_P in Eq (3); since θ_P and δ_{P1} appear only through the product $\kappa_P \equiv \delta_{P1}(1-\theta_P)$, the pair is structurally non-identifiable when both are unknown, and in the Normal regime ($\theta_P = 1$) δ_{P1} leaves the model entirely.

4.2 Experimental design and synthetic data

The base design has ten conditions per regime (Fig 1C): the treated baseline; five retinoid arms (no ATRA; early, late and low-dose ATRA; a strong-circadian arm without ATRA); and four arms that perturb the WNT/MYC sub-network through known inputs (a WNT pulse, a sustained WNT-inhibition step $D_W = -0.7$ from $t = 50$, a MYC pulse, and a combined WNT and ATRA arm). Reference trajectories are computed with the implicit Radau method at relative and absolute tolerances 10^{-10} and 10^{-12} [47]. Each condition is observed at the initial time and 149 further times drawn uniformly without replacement from a 5000-point grid on $[0, 150]$, all seven states, with independent Gaussian noise of standard deviation $\sigma = 0.002$. The benchmark dataset uses data seed 42 (seed $42 + c$ for condition c); a replicate dataset with seed 7, never used for any tuning, is used to test generalisation, and 24 further seeds (101–124) for the Monte Carlo study. Every estimator in Sections 2.3–2.6 sees byte-identical data.

4.3 Fisher information and the Cramér–Rao bound

All parameters are estimated in log coordinates, $x = \log(\theta/\theta_0)$. For the design above, with residuals $y_{cki}(x) - y^{\mathrm{obs}}_{cki}$ over conditions c, times k and states i, the Fisher information is

$$\mathscr{I}(x) = \sigma^{-2} J^\top J, \qquad J_{(cki),j} = \frac{\partial y_{cki}}{\partial x_j}, \tag{9}$$

computed at the true parameters by central differences (step 10^{-4} in log space) on Radau solutions at tolerance 10^{-10}. For an unbiased estimator, $\mathrm{Cov}(\hat{x}) \succeq \mathscr{I}^{-1}$, and the square roots of the diagonal of $\mathscr{I}^{-1}$ are the CRLB standard deviations; in log coordinates they are relative standard deviations. Because (θ_P, δ_{P1}) span an exact null direction of the 36-parameter $\mathscr{I}$ (Fig 3A,B), bounds are reported for the 35 parameters with δ_{P1} known. Fixing δ_{P1} removes the null direction and nothing else: the bounds of the other 34 parameters are the same whether the degenerate pair is represented by θ_P or by the identifiable product κ_P.

The PINN, the SGN-PINN and the reference classical fit minimise a relative-weighted least-squares objective in which state i carries weight $w_i = 1/\max(\max_t |y_i|, 0.05)$ (the weighting the PINN uses). The asymptotic covariance of that estimator is the sandwich $\sigma^2 (J^\top \Omega^2 J)^{-1} J^\top \Omega^4 J (J^\top \Omega^2 J)^{-1}$, with $\Omega = \mathrm{diag}(w_i)$, which is not the CRLB: its standard deviations exceed the bound by a median of 7–9% and at most 47% across parameters and regimes, and the

CRLB-expected count of the weighted estimator is lower by at most 0.3 parameters per regime. We use the CRLB itself, the tighter yardstick, throughout, and the unit-weight MLE wherever the question is whether the bound is attained.

The CRLB-expected count. A recovery count is the statistic every earlier study of this model reports, so we translate the bound into the same currency. If $\hat{x}_j \sim \mathcal{N}(x_j, s_j^2)$ with s_j the CRLB standard deviation, the probability that $\hat{\theta}_j$ is within 10% of the truth is $\Phi(\log 1.1/s_j) - \Phi(\log 0.9/s_j)$, and the sum over parameters, $\mathbb{E}[N_{10}]$, is the number an efficient estimator recovers on average. We report it over the 34 parameters outside the (θ_P, δ_{P1}) pair, and report θ_P separately.

4.4 Maximum-likelihood estimation

The classical reference estimator is single shooting: the ODE is integrated at each trial parameter vector (LSODA, relative tolerance 10^{-11}) and the weighted residual is minimised by a trust-region reflective least-squares method with forward-difference Jacobians (step 10^{-5} in log space) computed in parallel, run to convergence (relative tolerances 10^{-12}, at most 400 evaluations; one fit that reached the cap was rerun with 1500 and converged at 536, and two that reached it in the design comparison are either not selected or agree with the next start's cost to 10^{-8}) [47]. The solver tolerance matters: with LSODA at 10^{-8} and a step of 10^{-6} the finite-difference Jacobian is too noisy along the flattest directions, and fits in the two high-WNT regimes stop short of the optimum, so every fit reported here uses the tighter settings. Two weightings are used: the relative weights w_i above (the shared objective of all estimators) and unit weights, which make the estimator the exact maximum likelihood estimator (MLE) for the iid noise and are used for the Monte Carlo study. Starts are the common initial guess ($1.5\times$ truth, $W = 1.2$, $\theta_P = 0.6$, in every regime) and three starts with $\mathcal{N}(0, 0.35^2)$ log-jitter around it. The reported estimate of a multistart is the start with the lowest final cost; the truth is never used for selection.

Monte Carlo test of the bound. For each regime and each of 24 data seeds, the unit-weight MLE with δ_{P1} known is started at the truth and run to convergence; the empirical standard deviation of $\hat{x}_j$ across seeds is compared with the CRLB, and the standardised errors $z_j = (\hat{x}_j - x_j)/s_j$ with a standard normal. Starting at the truth isolates the statistical question (is the bound attained?) from the optimisation question (is the basin found?), which the multistart runs address.

4.5 Physics-informed neural networks

One network per condition represents the state trajectory, $z_\phi : t \mapsto \mathbb{R}^7$: a GELU multilayer perceptron of width 256 and depth 4 on a fixed Gaussian random Fourier embedding of time (16 frequencies, scale 4) [48, 49], with 207 879 weights per condition; the embedding is needed because a plain network's spectral bias [50] flattens the circadian forcing. A single parameter vector is shared by all conditions. The loss is the weighted data misfit, an initial-condition penalty and a physics residual on 8000 collocation points, balanced adaptively, and is minimised with Adam [51] (2000 epochs; learning rates 10^{-3} for the networks, 5×10^{-3} for the parameters) then L-BFGS [52], followed by a parameter refinement with the networks frozen. Three variants form a matched ablation, each run from one start on the benchmark data (the integral variant also on the replicate): *autodiff*, the standard residual $\dot{z}_\phi - F(z_\phi; \theta)$ with $\dot{z}_\phi$ from automatic differentiation, unit weights and random collocation; *weighted*, which adds the relative weights w_i and a fixed collocation grid; and *integral*, which replaces the residual by the derivative-free trapezoidal form

$$R_k = \frac{z_{k+1} - z_k}{\Delta t} - \tfrac{1}{2}\big[F(t_k, z_k; \theta) + F(t_{k+1}, z_{k+1}; \theta)\big], \tag{10}$$

which never differentiates the network in time. The forward PINN of Section 2.2 uses the same architecture with the parameters known, 40 noise-free observations per state, the initial condition and 50 000 collocation points.

4.6 The Schur-complement Gauss–Newton PINN

The SGN-PINN keeps the PINN's representation and changes how it is optimised. For each condition the state is linear in a head matrix A_c,

$$z_c(t) = \big(h(t)^\top A_c\big) \odot s_c, \qquad h(t) = \big[\, \psi_{\mathrm{PINN}}(t),\ \sin \omega_\ell t,\ \cos \omega_\ell t,\ e^{-t/\tau_m},\ 1 \,\big], \tag{11}$$

where ψ_{PINN} are the 256 last-hidden features of the trained integral PINN of that condition (frozen), ω_ℓ are 256 random frequencies uniform on $[0, \pi]$ (512 Fourier features), τ_m are 24 log-spaced decay times in $[0.05, 20]$ that represent the fast initial transients, and s_c is the PINN's output scale. With the features fixed, the unknowns are $(A_1, \ldots, A_{10}, x)$, $7 \times 793 \times 10 + 36 \approx 5.6 \times 10^4$ numbers, and the objective

$$\mathscr{L}_\lambda = \sum_c \left\| \frac{\Omega(z_c(t_d) - y_c)}{\sigma} \right\|^2 + \lambda \sum_c \Delta t \left\| \frac{\Omega R_c(A_c, x)}{\sigma} \right\|^2 + \lambda_0 \sum_c \left\| \frac{\Omega(z_c(0) - y_0)}{\sigma} \right\|^2 \tag{12}$$

is a nonlinear least-squares problem in which R_c is the residual (10) on a 3000-point grid and $\lambda_0 = 100$. Its Gauss–Newton matrix is block-arrowhead: ten dense head blocks $\mathscr{A}_c$, coupled only through the 36 parameters by blocks $\mathscr{B}_c$, and a parameter block $\mathscr{X}$. Each Levenberg–Marquardt step [53, 54] therefore costs ten Cholesky factorisations of 5551×5551 blocks and one 36×36 solve with the Schur complement

$$\mathscr{S} = \sum_c \left[\mathscr{X}_c - \mathscr{B}_c^\top (\mathscr{A}_c + D_c)^{-1} \mathscr{B}_c \right], \tag{13}$$

the reduced Gauss–Newton matrix of variable projection [29, 30]. The physics weight is continued upward, $\lambda = 10^{-2} \to 10^3$, from a smoothing problem in which the data dominate to one in which z_c must solve the ODE; at large λ the estimate approaches the relative-weighted least-squares estimate restricted to the span of h. The undamped Schur complement is then the Gauss–Newton curvature of the objective profiled over the heads; when the physics term is satisfied and the weights are close to uniform it approximates the profiled Fisher information of x, and we report $\mathrm{diag}(\mathscr{S}^{-1})^{1/2}$ as the estimator's own standard error, obtained at no extra cost, and test that approximation against the CRLB of the design (Section 2.6). Two schedules are run from the integral PINN's estimate (the reference formulation when the method was developed; the ablation of Section 2.5 was run afterwards): S1 ($\lambda = 10^{-2}, \ldots, 10^3$, ten iterations per level) and S2 (15 head-only Gauss–Newton iterations at fixed x, then $\lambda = 1, \ldots, 10^3$, twelve iterations per level). The reported estimate is the schedule with the lower *shooting* misfit, the weighted data misfit of the ODE solution at the estimated parameters, which requires no trust in the head and no knowledge of the truth. The basis and both schedules were fixed on the seed-42 data; the seed-7 dataset is held out.

4.7 Hybrid models, the equation-local screen and the anchor ratio

A hybrid replaces one closed-form regulation with a network. For the main example, the β-catenin→MYC induction,

$$\eta_{BM} H_2(b; \kappa_{BM}) \longrightarrow f_\phi(b), \qquad \varepsilon_M \dot{m} = a_M + f_\phi(b) - m, \tag{14}$$

which removes (η_{BM}, κ_{BM}) from the unknowns and leaves the basal production a_M, the decay and the timescale mechanistic. f_ϕ is a two-hidden-layer, five-unit tanh network (46 weights) with an L^2 penalty of 10^{-8} on the mean squared weight, a value calibrated to the scale of this objective, small, Glorot-initialised and multi-started as recommended in [11]. Thirteen edges of the model can be learned this way (S2 Table): seven *production* edges of the form $a + f(u)$, four *modulator* edges of the form $f(u) \cdot y$, one multivariate ratio, and the APC-loss law. Five constraint parameterisations are implemented for a production edge, all non-negative: `gated`, $f_\phi(u) = \mathrm{softplus}(g(u))(1 - e^{-u/u_0})$ with $u_0 = 0.1\, u_{\max}$ (exact anchor, not monotone); `sc`, $\mathrm{softplus}(h(u)) - \mathrm{softplus}(h(0))$ with h monotone through non-negative weights (an exact anchor); `sc_bounded`, as `sc` with the true asymptote supplied as a bound (an oracle no real analysis would have); `lin`, $(u/u_{\max})\,\mathrm{softplus}(g(u))$; and `lin_mono`, the same with a monotone rate. `sc` follows the monotone construction of [26] with an exact anchor added.

The full hybrid pipeline (all state networks, all conditions, all parameters) is used for the round-one and confirmation results. Every other hybrid experiment uses an equation-local screen: only the equation hosting the learned edge is fitted, with the noise-free reference states supplied on a fixed grid, every parameter outside that equation held at truth, and the equation's own parameters started from the same $1.5\times$ guess; the residual is (10) on that one row. Each cell is fitted with the network and, from the same seeds and budget, as a mechanistic control with the closed form and all of its parameters unknown (3000 Adam steps, 300 L-BFGS iterations, the restart with the lowest residual kept). Because exact states go in, the screen excludes state-estimation error and its absolute errors bound what the full pipeline achieves from below. Restart counts differ between experiments and are stated with each result; selection by lowest residual is not always selection by lowest parameter error, so numbers are compared within an experiment, not across experiments.

For a regulator u under a condition set, the anchor ratio

$$\rho_u = \frac{\min_{c,t} u_c(t)}{\max_{c,t} u_c(t)} \tag{15}$$

is the fraction of the regulator's observed range that separates the data from the anchor $u = 0$; it is computed from reference trajectories alone.

4.8 Fisher information of a learned term

For the hosting equation k of a production edge, the screen minimises the trapezoidal residual of that one equation, with the reference states supplied on a fixed grid (stride 8 on the 5000-point reference grid), scaled per condition

by $s_c = \max(\max_t |z_k|, 0.05)$. Reading that objective as a Gaussian likelihood with residual noise σ_r gives a Fisher information $\mathscr{I} = \sigma_r^{-2} J^\top J$ over the equation's mechanistic parameters (in log coordinates) and the learned term's weights. The learned term is represented in three ways: by the closed-form Hill term (the mechanistic control); by the screen's own `gated` and `sc` networks, fitted to the true Hill term (median over three fits); and by a linear family $\sum_{k \le K} c_k (u/u_{\max})^k$ that satisfies the anchor exactly. The network block is profiled out, $\mathscr{I}_{a|\phi} = J_a^\top (I - P_\phi) J_a$, with P_ϕ the projector on the network columns of J above a relative singular value tolerance 10^{-8}; the reported quantity is $\Gamma = \mathrm{sd}(\log a)/\sigma_r$, which is free of the noise scale. Away from the anchor the network block is nearly rank-deficient and Γ depends on that tolerance (by a factor of 17 to 8000 between 10^{-6} and 10^{-10} under the base design, depending on the representation); it does not when the anchor is visited, and every rank correlation we report changes by at most 0.04 across the three tolerances. Jacobians are computed by automatic differentiation and checked against central differences; the bound is checked against 200 Monte Carlo refits with simulated residual noise for the MYC edge under two designs and two representations. Designs are the base design, the depletion and information-matched pairs, the protocol ladder of the design table and the dose arms, each at the true parameters. The analysis needs no training and runs on eight CPU cores in 13 minutes for all 29 designs.

4.9 Scoring and reporting

A parameter is recovered if its estimate is within 10% relative error of the truth; N_{10} is the number recovered. Counts are reported per regime and never pooled across different denominators. The functional error of a learned term is $100 \cdot \mathrm{RMS}(f_\phi - f_{\mathrm{true}})/\mathrm{RMS}(f_{\mathrm{true}})$ over the observed input range. Predictions for the dose-response and the prospective test were written down before the corresponding fits were run and are reported as scored (S1 Text). Computation used one NVIDIA GPU with 95 GB and a 64-core CPU node; measured wall-clock times are compared in Fig 6E.

Data and code availability

All data are synthetic and are regenerated by the code from fixed seeds. The model, every estimator, every experiment script, the result files (JSON) behind every number, and the figure scripts (each writing an audit file that lists every plotted number with the run artifact it comes from) are available at `https://github.com/aidxhxr/PINN-Research/tree/paper-v1`, the tagged state from which this paper was built (paper sources in `research-paper-final/`, with `REPRODUCE.md` linking every figure and every number quoted only in the text to its script and run).

Author contributions

Amirkhan Aidarkhan: conceptualization, methodology, software, investigation, formal analysis, visualization, writing (original draft). **Pascal K. Kataboh:** conceptualization, supervision, writing (review and editing).

Competing interests

The authors declare no competing interests.

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Supporting information

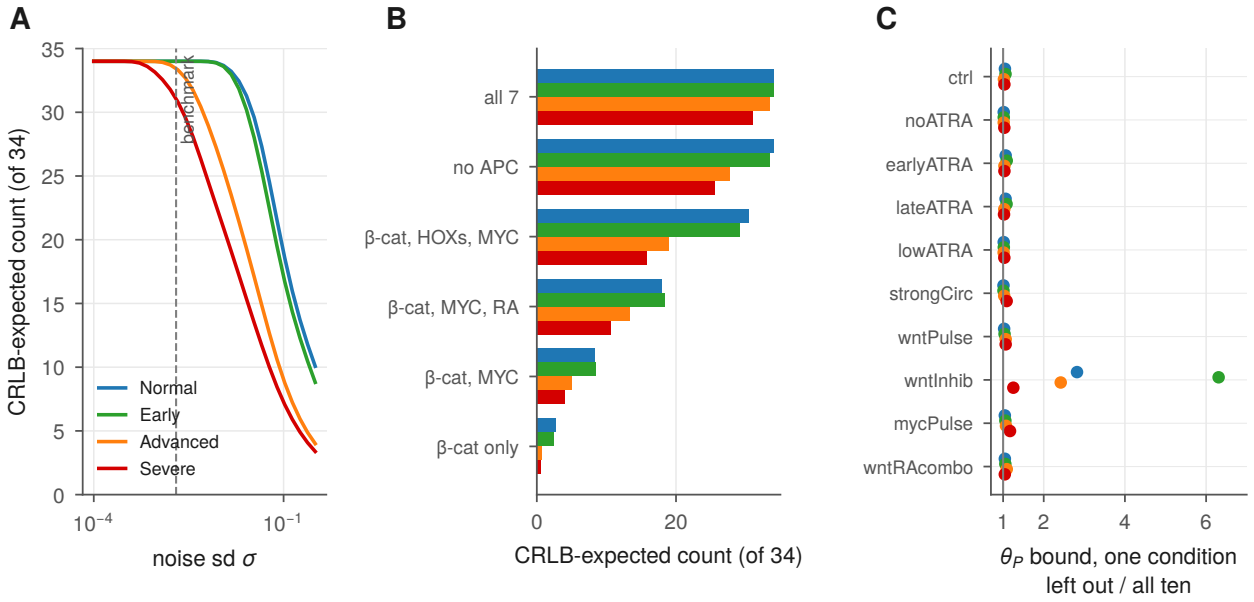


Fig S1. S1 Fig. How the information budget depends on the design. From the per-condition sensitivities of the benchmark design at the truth. (A) CRLB-expected number of the 34 regular parameters recovered within 10% against the observation-noise standard deviation (all seven states observed; dashed, the benchmark's $\sigma = 0.002$). (B) The same at $\sigma = 0.002$ when only a subset of states is observed. (C) Bound on θ_P (δ_{P1} known) when one of the ten base conditions is left out, relative to the full design: only the sustained WNT-inhibition step matters.

Table S1. S1 Table. Parameters. The 36 kinetic parameters, unknown in every inverse problem, with their baseline values, and the known forcing parameters. Initial state $(b, p, h_5, h_{13}, m, r, c) = (0.20, 1.00, 0.80, 0.30, 0.30, 0.60, 0.40)$. W and θ_P take the regime values of Table 1.

equation	parameters (value)
b	W (regime), η_{13} 0.75, κ_{13} 0.55, λ_P 1.60, λ_5 1.30, κ_5 0.50
p	θ_P (regime), ε_P 1.00, ρ_5 1.10, ρ_B 1.10, ρ_{13} 1.30, δ_{P1} 3.50
h_5	ε_5 1.20, a_5 0.15, η_R 2.50, κ_R 0.40, η_M 2.50, κ_M 0.50
h_{13}	ε_{13} 1.00, a_{13} 0.18, η_{B13} 0.95, κ_{B13} 0.50, η_{M13} 0.55, κ_{M13} 0.50
m	ε_M 0.60, a_M 0.18, η_{BM} 1.35, κ_{BM} 0.50
r	ε_R 0.40, λ_C 0.85
c	ε_C 0.80, a_C 0.08, η_{RC} 1.50, κ_{RC} 0.50, η_{BC} 1.50, κ_{BC} 0.50
forcing (known)	μ_0 0.35, A_R 0.04, T_R 24, φ 0, D_R 1.50, q 0.30, (τ_1, τ_2) (40, 88)

S1 Text. Pre-registered predictions and their scoring

Dose-response (registered before any fit). P1: within each arm and regime, the excess basal error (hybrid minus control) is monotone in achieved anchor ratio (Spearman ≥ 0.8). P2: the retinoid arm falls below 2% at $k = 0$ and below 3% by $k = 0.03$. P3: the single-knockdown arm never falls below 2% at any dose (the registered falsifier: a collapse would mean knockdown depth, not anchor visitation, is doing the work). P4: the double-knockdown arm at $k = 0$ is below 2% and below the single-knockdown arm at $k = 0$. P5: the retinoid and double-knockdown arms lie on one curve against ratio. P6: matched ratios reached by different means give errors within a factor of about 2. Scoring with the best available data (Normal two restarts, Severe five): P1 fails (four of six arm–regime cells pass at $\rho_s = 0.94$–1.00; `wnt`/Normal +0.58 and `ra`/Severe +0.26 fail); P2 passes in Normal (0.0%, 0.2%) and fails in Severe (3.3% against a mechanistic-control floor of 3.4%); P3 fails in Normal (1.3%) and passes in Severe (2.5%); P4, P5 and P6 pass in both regimes (P5 gap 0.4 and 3.3 points; P6 $\times 2.0$, $\times 2.3$). Three of six pass in both regimes. With two rather than five restarts in Severe, P3 and P5 also failed there: P3 because the single knockdown reached 1.5% at $k = 0$ (2.6% with five restarts), P5 because of one bad optimum at $k = 0.1$ (23.4% between neighbours at 0.8% and 3.3%), which five restarts removed.

Table S2. S2 Table. The thirteen learnable edges. A production edge has a basal parameter beside the learned function and is exposed to the compensation identity, Eq (1); a modulator edge multiplies a state and is not.

edge	class	equation	input	replaces	basal
`ra_h5`	production	h_5	r	$\eta_R H_1(r;\kappa_R)$	a_5
`bm_myc`	production	m	b	$\eta_{BM} H_2(b;\kappa_{BM})$	a_M
`b_h13`	production	h_{13}	b	$\eta_{B13} H_2(b;\kappa_{B13})$	a_{13}
`m_h13`	production	h_{13}	m	$\eta_{M13} H_2(m;\kappa_{M13})$	a_{13}
`bc_cyp`	production	c	b	$\eta_{BC} H_2(b;\kappa_{BC})$	a_C
`rc_cyp`	production	c	r	$\eta_{RC} H_1(r;\kappa_{RC})$	a_C
`h13_b`	production	b	h_{13}	$\eta_{13} H_2(h_{13};\kappa_{13})$	W
`m_h5`	modulator	h_5	m	$\eta_M m/(\kappa_M+m)$, times h_5	–
`h5_b`	modulator	b	b	$\lambda_5/(\kappa_5+b)$, times $h_5 b$	–
`apc_b`	modulator	b	p	λ_P, times $p\,b$	–
`c_ra`	modulator	r	c	λ_C, times $c\,r$	–
`apc_prod`	ratio	p	(h_5, b, h_{13})	APC production ratio	–
`apc_mutation`	parameter	p	$1-\theta_P$	$\delta_{P1}(1-\theta_P)$	–

Table S3. S3 Table. Full-pipeline confirmation. Full inverse PINN with two depletion conditions added to the ten, three restarts, counts over the 34 shared parameters, each hybrid against its own mechanistic control on the same twelve conditions. Upper blocks: retinoid-free medium plus the single WNT knockdown ($k_W = 0.02$); lower block: retinoid-free medium plus the β-catenin double knockdown ($k_W = k_{13} = 0$). gate_{lo} is the gated network's gate at the lowest observed input (0: the anchor binds). Runs `PINN-hybrid-ude/runs/20260802_041545_ra_h5_dep`, `20260802_010526_bm_myc_dep`, control `20260801_211530_control_dep`; double knockdown `20260921_184115_bm_myc_depb`, control `20260921_172342_control_depb`.

hybrid	regime	recovered (control)	term NRMSE	basal error (control)	gate_{lo}
RA→HOXA5	Normal	18 (18)	11.6%	13.0% (1.7%)	0.00
	Early	20 (17)	9.6%	4.7% (14.0%)	0.00
	Advanced	18 (14)	9.5%	2.4% (19.5%)	0.00
	Severe	18 (14)	9.6%	3.8% (17.9%)	0.00
β-catenin→MYC, single KD	Normal	15 (17)	2.3%	10.3% (0.4%)	0.14
	Early	16 (16)	2.4%	11.7% (0.2%)	0.14
	Advanced	13 (13)	2.9%	17.4% (0.4%)	0.13
	Severe	13 (13)	2.7%	16.4% (0.5%)	0.12
β-catenin→MYC, double KD	Normal	18 (19)	1.8%	1.4% (0.3%)	0.00
	Early	16 (17)	1.2%	0.6% (0.2%)	0.00
	Advanced	13 (15)	4.4%	0.6% (0.2%)	0.00
	Severe	11 (11)	4.2%	0.7% (0.3%)	0.00

Table S4. S4 Table. The design table. Anchor ratio of each regulator, worst over the four regimes, under each protocol added to the ten base conditions, ordered by laboratory cost; computed from reference trajectories without training. `mycKO` is the β-catenin double knockdown plus silencing of basal MYC transcription ($k_{aM} = 0$); `hox13KO` adds $k_{a13} = 0$ and therefore also contains $k_W = 0$. APC never falls below 9.5% of its range: its production numerator $1 + \rho_5 h_5$ is bounded below by one. Run `PINN-hybrid-ude/runs/20260802_anchor_reach`.

protocol	ρ_p (APC)	ρ_b	ρ_c	$\rho_{h_{13}}$	ρ_m	ρ_r
ten base conditions	0.179	0.081	0.181	0.193	0.116	0.143
retinoid-free medium	0.163	0.081	0.181	0.193	0.116	**0.000**
WNT knockout ($k_W = 0$)	0.174	0.013	0.181	0.163	0.079	0.136
β-catenin double KD ($k_W = k_{13} = 0$)	0.171	**0.000**	0.181	0.161	0.079	0.136
`mycKO`	0.117	**0.000**	0.181	0.119	**0.000**	0.136
`hox13KO`	0.095	**0.000**	0.181	**0.000**	**0.000**	0.136
`cypKO`	0.179	**0.000**	**0.000**	0.161	0.079	**0.000**

Prospective test (registered before any fit). Q1: `m_h13` basal error below 3% under the prescribed protocol (`mycKO`). Q2: under the near-miss `bcatKO` it stays above 8% in at least one regime (the falsifier). Q3: the equation-parameter count rises under `mycKO` and not under `bcatKO`. Q4: the functional error under `mycKO` falls below 20% in Normal. Q5 (secondary, flagged in advance as confounded): `h13_b` basal error below 5% under `hox13KO`. Scoring: Q1 passes in both regimes; Q2 passes; Q3 passes in Normal and fails in Severe; Q4 fails (68.4%); Q5 fails in both regimes. Two of

five pass.

S2 Text. Training and computational details

Inverse PINN. One state network per condition (Methods); 150 observations per condition with Gaussian noise of standard deviation 0.002, plus the initial condition (weight 20); 8000 collocation points; Adam (10^{-3} for the networks, 5×10^{-3} for the log-parameters, 2000 epochs, cosine schedule), then 150 L-BFGS steps and 600 parameter-refinement steps with the networks frozen; adaptive loss weights. Parameters are initialised at $1.5\times$ truth with $W = 1.2$, $\theta_P = 0.6$ in every regime.

SGN-PINN. Head features as in Eq (11); 3000-point residual grid; initial-condition weight $\lambda_0 = 100$; Levenberg–Marquardt damping initialised at 10^{-3}, divided by 3 on acceptance and multiplied by 5 on rejection, ridge 10^{-10} times the mean data-block curvature on the head blocks; a level of the λ schedule ends after the iteration budget, a relative decrease below 10^{-7}, or eight consecutive rejections. All linear algebra in double precision on the GPU.

Hybrid term and screen. 2×5 tanh network, Glorot-uniform, final bias zero, L^2 penalty 10^{-8} on the mean squared weight, trained jointly. Screen: stride-8 reference grid, 3000 Adam steps at 5×10^{-3}, 300 L-BFGS iterations with strong-Wolfe line search, restart seeds $100 + 17s$, lowest residual kept.

Fisher information. Central differences with step 10^{-4} in log-parameters on Radau solutions (rtol 10^{-10}, atol 10^{-12}), at the 150 observation times of each condition of the benchmark dataset; $\sigma = 0.002$. The design-comparison bounds for θ_P (Section 2.7) use 200 uniformly spaced times per condition and a step of 10^{-3}.